

ActInf GuestStream 064.1 ~ Elliot Murphy "ROSE: A neurocomputational architecture for syntax"

GuestStream | Elliot Murphy | 2:46:03

Hello and welcome. This is active guest stream 64.1. It's November 29th, 2023. We're here with Elliot Murphy, returning guest, and we'll be talking about this paper, Rose, a neurocomputational architecture for syntax. So for introduction and describing how today's stream is going to go, Elliot, thank you for joining. All right. Thanks so much, Daniel. Yeah, I work at the UT Health in Houston in Texas in the epilepsy monitoring unit where we use intracranial recordings to map language in patients with epilepsy. But today I want to present something a little bit more theoretical, a little bit more kind of philosophical almost. This is a new paper that's just come out in the Journal of Neurolinguistics, and I thought it'd be good for me just to kind of walk through the paper step by step, give some motivation, give some description, give some kind of outline of the general kind of framing for the paper. So the framework for this paper is trying to propose a general model for how to research language using intracranial recordings. It's basically trying to move beyond research that comes from other kinds of extracranial imaging methods, and it's trying to really figure out what is the most appropriate level of description in terms of neural complexity for specific aspects of language. So there's many different parts of language, as we know. Language has words, it has sounds, you know, it has, you can read language, you can sign it, you can construct meaning, you can do all sorts of things. But the real kind of question for me is, at what level of neural complexity, and therefore at what level of appropriate recording resolution, based on what tools you have available, is going to be providing you with the most reliable signatures of particular levels of structure. So in this model, the Rose model, I invoke representations, operations, structures, and encoding, and I'll introduce them step by step. And so for the purposes of this kind of introduction, I'm just going to kind of outline what those general kind of frameworks are. But to begin, the most kind of basic assumption that we have about language, in terms of like the findings from linguistics, so the study of language, like the scientific study of language, one of the really basic introductions that you find in every introductory textbook, is that linguists have kind of arrived at the conclusion that the human mind brain applies a set of rules to recursively combine linguistic units into larger objects, which derives an unbounded array of hierarchically structured expressions, with humans in varying sentence meaning based on syntactic configuration. Now that's a lot of jargon. But essentially, what that means is, when you hear a linear sequence of words, like, you know, we watched a movie with Jim Carrey, you can pass that not simply as like beads on a string, you can pass it in two ways, depending on how you chunk the individual phrases together, and how you establish relations between different elements. So for example, you know, we watched a movie with Jim Carrey can mean that you watched, you know, Eternal Sunshine of Spotless Mind, or it can mean that

you're sat there next to Jim Carrey watching, you know, the new Napoleon film, for example. So there's this kind of ambiguity there, but the ambiguity arises due to structural configurations. So that's kind of how you rearrange the sentences kind of in your head internally. And that is obviously a process that's independent immediately from sensory motor, you know, transformations or experience or speech or sound or orthography. So these are very abstract processes. And they're clearly mental processes. They're not processes of the visual system. They're not processes of the auditory system. They're clearly high order mental computations. So what that phrase, unbounded array of hierarchically structured expressions means, is it just means that humans have this generative capacity. And people like Carl Christen and Chomsky talk about generativity quite a lot. What it basically means is the ability to just do exactly what I'm doing right now, use a finite set of elements. So lexical items, or, you know, memorize words, and then recombine them again, again, again, and really novel combinations, such that you can say sentences that have, you know, never been said before, in, you know, the totality of human existence, but which are immediately assigned a meaning. You don't have to think too hard about what meanings are usually, unless you're reading, you know, *Finnegan's Wake*, for example. So typically, this is a reflexive process. So it's automatic, it's quick, the brain resolves this very well, without much effort, without much energetic cost as well. And so the question is, how does the brain do this? And so, you know, an additional point to make here is that this capacity for hierarchical occasion has also been linked to human specific cognitive superiority. So that's a more general background here. So people like Stan Dehane, Hauser, and Wattenwald, they've long stressed that this facility for constructing hierarchically organized linguistic structures is really potentially a human specific function. And that's a very long and, you know, it's a long and difficult topic to get to get into. So I'll put it to one side. But it's an interesting topic to explore the possibility that this whatever this process is, it might in fact be unique to humans.

So in the theoretical domain, and turning to neurobiology, an emerging consensus in neuroscience is that complex behavior and cognition rely on coordinated interactions between brain regions, with phase synchronization being a major candidate for a mechanism implementing this coordination by gating information transmission in the brain. But unlike for models of attention and working memory, there's currently an absence of oscillatory phase coding in models of natural language. So what does that mean? So in the neurobiology of attention and working memory, there's been a lot of really exciting accounts by people like L. Miller, Oli Enson, and a lot of other people trying to explore the relationship between neural synchronization metrics, and some kind of isomorphic kind of relationship to some property of an endogenous computation, like a working memory process, or some kind of property of the environment that you're kind of, you know, passing in real time. And but in language, the best we have just to boil it down very, very generally, the best we have at the moment in language is basically cartographic models. So that's kind of localization as models in the brain. So we have a decent idea, thanks to the really brilliant work by people like Greg Hickok, and William Machen, and Angler Friedrichi, and Niding and David Popol, and really,

you know, a really incredible roster of people from the last few decades, who very carefully mapped out which specific brain regions seem to be sensitive to different types of linguistic structure. So lexicality, meaning, you know, the meaning of a word, and syntax, the generation of a sentence structure, a phrase structure, semantics, the meaning of words, phonology, and so on. But what we really have right now is just a kind of map. So we have a map of the brain that you can pull up on Google, and you can Google, you know, language syntax brain, and you'll get a decent, you know, rough architecture of which parts of the brain seem to be sensitive to different parts of language. But the main reason why I wrote this paper is because that's essential, you know, mapping out the terrain is obviously essential. But the next step, people from working memory, vision, attention, they did this, you know, a while ago, the next step is therefore to transition to a kind of how question, which is which neural mechanisms are involved in language. So rather than simply saying, these are the brain regions involved, we have to then say, okay, these are the brain regions, but what is the brain doing, right? What operations are it performing? And at what level of neural complexity? And is it sensitive? So just because this part of the brain is sensitive, we don't really know in what way it's sensitive, like how is it what type of neural processes is it actually recruiting to yield that sensitivity? And what does that sensitivity really look like outside of all fluctuations or, you know, a high gamma amplitude increases and other kind of difficult measurements? So that's in the theoretical domain. In the empirical domain, invasive intracranial recordings have recently mapped out some of the feature space associated with how much phonological information can be transmitted by different neural mechanisms and some citations there for you to look at. But within the domains of syntax and semantics, there's been less clear progress. And the reason for that is pretty obvious. Phonology, sound, auditory processing, you know, the kind of finite feature space involved in articulatory processes, that's a very non-trivial, you know, field to explore. But it has been mapped out very clearly because the, you know, processes are kind of less abstract, more well agreed upon, more transparent. They involve more clear sensory motor transformations and things that you can detect very clearly with very nice clear signatures in different types of neural states, state space, state space, or in fact, different regions of the brain. And these are also things that you can disrupt very easily. So it's very easy to disrupt phonological processing. We know which parts of the brains, which parts of cortex is disrupted. But it's less clear how you specifically disrupt things like syntax and semantics. You can knock out, you know, language comprehension in general, just all language comprehension. You can knock out and disrupt certain aspects of working memory performance. But when it comes to knocking out, you know, disrupting the brain's ability to specifically process syntactical semantic information, that's been very difficult. And there's only been a few papers on that, which I cite in this paper over the last couple of years. But it's very difficult to really try and, you know, provide causal evidence to try and really stimulate, stimulate the brain in the OR or to find, you know, lesion patients who've had a stroke,

and who exhibit extremely specific niche, you know, deficits around this aspect of syntax or semantics. So that's, that's kind of the obstacle that we have, right? The obstacle we have is that we have this very strange, potentially human specific, seemingly very simple, you know, process of just building hierarchical structures. But it's very difficult to isolate from all these other things going on. Because as soon as you build a sentence structure, as soon as you build a phrase or a sentence, you are also doing loads of other things. You are also using your attention, you're also using working memory, you're also saying things or reading things or engaging your eyes or ears. So that's, you know, you know, it really is very difficult to isolate it. And so anyway, that's, that's, that's the problem that we have. So I'll just skip this part to the next section here. So, and I'm just going to read through some of this text and maybe try and unpack it a little bit more. And but the past decade has seen the emergence of low frequency phase coherence as a feasible index of hierarchical syntactic structure, William. So what that means is that if you read these papers, and there's low frequency oscillations in the brain that can be detected using MEG, or scalp EEG. And they basically show little fluctuations, little peaks, aligned with certain moments and periods of or windows of structure building, I'm just thinking of more metaphor, little little little islands of structure that come along when you pass a sentence in real time, whenever that moment of structure building occurs, you seem to get this low frequency peak in activity in the brain. And it tends to be around, you know, classical language cortex, like left, left inferior frontal cortex or temporal cortex. But the isolate of the actual spatial temporal isolation is still a little bit iffy, we still don't really know, quite frankly. And so, so that's the kind of, you know, background for that aspect of language. At the same time, though, recent work has also examined local cortical processing using intracranial recordings. So this is these are recordings inside the brain. And what they do is they often focus instead on high frequency power, they can detect low frequency power, absolutely. But most of the analyses and novel findings within the intracranial space have prioritized looking into high frequency research. And for the simple reason that that's doing the real way that you can get high frequency recordings. And so this works expose signatures of syntax in other kind of areas outside of low frequency dynamics. So there's a big question here, how these two distinct recording scales, low frequency, you know, dynamics and high frequency dynamics could be combined in a coherent model of natural language syntax and cortical computation. And that hasn't been addressed yet. So that's kind of

one of the arguments that I try and build here. And so yeah, neural oscillations are effectively, like I mentioned, they reflect synchronized fluctuations in neuronal excitability, and are typically grouped by frequency and mainly for convenience. But you know, not always, there are in fact some, there seems to be something else going on. And common rhythms are often termed delta, theta, alpha, beta, and then gamma. And broadly speaking, high frequency activity reflects local neuronal processing, whereas lower frequency activity reflects regional synchronization. So high frequency activity, you get that local cortical computation in a specific brain region, this brain region is doing something, as opposed to low frequency activity, which is often more

widespread. And it's all it's more to do with the coordination of regional synchronization. So the low frequencies say, okay, it's time for you guys to all come online at this particular time, to, you know, engage your spike activity, in this particular time period, but over a larger portion of tissue typically, it's over a much larger areas of the brain. And, and then there's a bunch of other kind of computational differences, seemingly as well between these lower bands. So for example, alpha is very often implicated in cortical disinhibition, it kind of helps to shield and protect ongoing representations from, you know, decay and transformation. And then theta oscillations, often implicated in hippocampus, have been involved in learning and memory, there's a big literature on that. Don't need to go into the details. But essentially, there's a whole, you know, panoply of low frequency rhythms, and they seem to be sensitive to different cognitive processes, or at least, you know, recruited, or they seem to index a bunch of lower order processes, potentially.

Okay. And so one way to kind of introduce this is through a model that I proposed in my book in 2020. And in this model, again, using invoking low frequency power, the combinatorial power of language is indexed via various oscillatory interactions, like forms of cross frequency coupling, which is basically when the phase of a lower frequency rhythm kind of coordinates and dictates the firing of a higher frequency rhythm. So you kind of get a potential facilitation of what's called multiplexing, or kind of packaging representations in one order, and then sending them downstream and kind of re-accessing them from in a different part of the brain. So in this book of mine, there's a bunch of kind of literature that's reviewed to motivate the idea that delta theta phase amplitude coupling constructs multiple sets of syntactic and semantic features. So you get this initial low frequency phase interaction between delta and theta. And this occurs when the phase of delta is synchronized with the amplitude of theta. And delta represents superordinate syntactic categories, whereas theta represents feature bundles generated via lexical access. What does that mean? It means that these theta oscillations, there's something called theta gamma coupling, famous in memory research and navigation. Theta gamma coupling, and in fact, there's some really nice work published yesterday, I think by Oli Enson, looking at how attention might be also implemented through alpha gamma coupling. But the basic idea is that you have individual sets of features that are, you know, basic mental representations, like things responsible for, you know, animacy, plurality, basic conceptual categories, they seem to be triggered by high frequency activity. And then they're coordinated and packaged into a single kind of chunk by theta. So you have a single low frequency theta wave. And within that single wave, you get a bunch of individual, you know, five, six, seven, or eight, however many you can kind of squeeze and fit in. And there's kind of trade off between fidelity and the kind of number of gamma oscillations you can package inside a theta cycle. But so that packaging of theta with gamma helps you to kind of coordinate and unify an individual representation in the, you know, attention and visual spaces that's been referred to as, you know, coordinated visual attention when you're kind of, your visual, your visual system is kind of attuning to particular properties in the environment. And in the language space, I'm kind of invoking it to unify different individual features that compose into a lexical item. So a lexical item is just a fancy word for a word. So something like,

you know, WAP talk, you know, bed, I'm just thinking of random words. All of these random words are obviously composed of individual features, conceptual features, right? So a bed is inanimate. It's, you know, has a particular geometry, it has a particular function, a purpose, a design, and so on. And all these different conceptual features compose into what we mean by a bed. Whenever you think of a bed or a chair or a mirror, or, you know, whole foods, it could be anything, it's simple, anything from simple to complex, you have a constellation of conceptual features that are being composed somehow. And the idea here is that these low-frequency theta oscillations are there to coordinate the firings of those individual features, which is why you get a lot of lexicality effects being found in, for example, the theta range. And then what you do is, in this model, you get those theta bundles of lexical features, and you embed them further. So you nest them again. And what you do when you nest them again within a delta rhythm is that the delta rhythm then takes over from the individual lexical items, and it composes a syntactic identity, so a structure, a structural identity. So for example, it's been, you know, well studied in linguistics that when you put two items together, two words together, you get a kind of symmetry breaking operation, where one of the items kind of determines the category of the phrase. So for example, if you say red boat, a red boat is a boat that is red. A red boat is not a red-like quality that has, you know, boat-like features, for example. So there's a kind of immediate asymmetry. And you get that all over the place. If you say John ran, John ran does not mean that there's a special kind of John who is exhibiting a running feature. It means that there's an event of running, and John was its agent, right? So there's always an immediate kind of asymmetry in language. And that's, you know, people like Stan DeHene and others have kind of speculated that one of the interesting things about language is that, unlike music and mathematics, that kind of allow symmetry in their structures, language doesn't really like symmetry. As soon as you get symmetry in language, the language system tries to break it somehow. Because there has to be some kind of categorizer for you to identify what's called the label or the category of the phrase, for you to get a single unified syntactic representation. So in other words, whenever you get two words being combined together, one of them always wins, and one of them's the loser. And that's how you build syntactic structures, you get a kind of categorization process. So just to kind of recap, you have delta, theta and gamma, but delta assigns or it somehow derives or facilitates the syntactic inference of what the category is. And the reason, who knows, right? I'll try and explain some potential reasons here. And ultimately, it's unknown. But we do know that in the empirical domain, we get a lot of these syntax signatures being found in the delta range. So delta seems to care about chunking on some level, maybe not syntactic specific chunking, but I think it's possible for reasons cited in the paper. So delta and then and train then packages within one of its waves, multiple theta rhythms. And the reason why is because the same mechanism that you invoke when you get theta gamma coupling. So it's a series of nesting, you know, nesting and nesting and nesting. And, you know, the whole rose architecture in this paper is basically trying to construct what some people might think of as rather crude, but I think rather empirically motivated, in fact, and isomorphic relationship between structure and

neural kind of signal complexity, right? It's it's common for people to think, well, you know, the brain's very complicated, language is very complicated, it surely can't be the case that a particular level of linguistic structure maps directly onto a particular resolution of, for example, oscillatory phase coding. And I think there's good empirical evidence to actually assume that and not everything takes place at the lower core screen level at the spike level, for example, or even intracellular. I think it's unlikely that all of these complex processes from making inferences about, you know, whether or not red or about wins and becomes the categoriser of the genetic phrase is all taking place at the intra or intercellular level, and at the kind of global, you know, the kind of stuff that you can detect using single unit recordings, for example. And I think something more global, and, you know, interactive is happening. And that's why you kind of need to invoke some kind of phase code, because the phase code, you only get that once you transition to more global synchronization metrics of the kind found with low frequency dynamics, that's the whole point of low frequency

synchronization. So I think that's kind of a kind of basic natural progression that you find here, at least in the way that I've motivated it in using the literature that I review in the book, and kind of is recapitulated here to a certain extent. Okay, so then we turn to something called phase resetting. So phase resetting, alongside concurrent encoding and storage of the products of this, these generated objects, permits a facility for recursive self-call. And so that's kind of where you get the recursive mechanism of language, you are consistently re nesting objects into this more complex delta rhythm, and chunking out more and more and more lower frequency, and complexes, essentially.

And just on the side, we also know that, from empirical work by people in my lab, and in the Netherlands as well, that you get a bunch of different dynamics alongside this. So for example, we know that beta rhythms, and other types of gamma rhythms are involved in all sorts of different types of processes, like conceptual storage, and maintenance of a particular structure and memory. And so there's a very kind of complex interplay here between different rhythms, and it's not intuitive, it's not easy to grasp on first, you know, I realize I'm not potentially not doing the best job of explaining it, because there's a lot of things that unpack, as soon as you move to another kind of concept, you have to unpack it to a certain extent. But the basic idea is that there's a lot of things that unpack, as soon as you move to another kind of concept, you have to unpack it to a certain extent. But the basic idea is that there's a lot of things that you have to unpack it.

To a certain extent, but the basic idea is that there's a number of interacting low frequency rhythms, that then dictate the firing rate and coordination of these faster rhythms, and the faster rhythms kind of are responsible for the more, the kind of smaller, more atomic features, like individual lexical items, and the slower rhythms are responsible for the more complex, larger structures, like syntactic identities, and syntactic workspaces, and things that you construct. So the kind of output, I guess, the output of the syntactic

generative engine is found at the lower frequency end, whereas the input, the kind of stuff that you need to provide to the system to get any syntactic structure building done in the first place, like for example, lexical features, you need at least to access some features, before you can combine them, in order to combine something, you need to have something to combine. Those individual properties are hosted by the high frequency oscillations, essentially.

Okay. And then finally, travelling oscillations are also going to be relevant here. So these are essentially, these low frequencies that I mentioned, except they exhibit a particular directionality, they migrate in phase coherence across the brain. And what that means is, they coordinate the spiking of neural clusters across the brain, across fixed points. So they kind of move from one part of the brain to another part. And as they do so, as they've, the phase of the low frequency is shifted across a particular path,

shifted across a particular path, they then trigger particular spiking profiles of these neurons that they travel across. So it's a coordination process. And it's also, of course, given the nature of a migration path, that naturally leads to the conclusion that, as is often commonly known, different parts of the brain are innately sensitive to particular representation of features of the environment, particular conceptual features, things like faces, you know, places,

orthographic information, and orthographic information, and emotions and things like that. There's a very big difference, for example, between inferior parietal cortex and anterior temporal cortex in terms of which conceptual features they seem to be sensitive to. So if you get that travelling profile, you get that low frequency, you know, phase code moving across a particular extended path, then that immediately gets you essentially for free, it gives you for free, the mechanism of activating distinct conceptual

representations during that process. So I think, you know, there's a lot of other things to say here, but travelling waves have been implicated in a number of kind of behaviourally relevant processes in humans and also nonhumans. And they seem to support brain connectivity and function more generally.

So yeah, that's kind of the background for travelling wave mechanism. And then finally, one final thing to say here, I mentioned the delta, theta, and gamma complex, there's also been some interesting work outside of language, looking at delta gamma feature combinatorics.

So that's kind of skipping out the middleman and simply involving this low frequency delta complex, and being phase coupled with the amplitude of high frequency gamma oscillations.

And what that seems to be involved in is feature combinatorics in general. So for example, it's been involved in fluid intelligence. So when you have to just perform a particular task or solve a problem immediately, and call upon various cognitive representations from different core knowledge systems in your mind, lots of different, you know, aspects of intelligence, you have to

kind of recruit here and there, you don't necessarily have to, in fact, you don't at all construct a hierarchical structure, you don't construct a hierarchical recursive representation, you just combine things together into a set into a conglomerate to help you solve a particular puzzle.

So that's what fluid intelligence is all about. You also get that with, say, in early stages in phrase composition in scale PEG. So people like Andre Martin and Jonathan Brennan have done some nice work showing this.

So what that tells you is that there's, you know, in language and also non-language, you can combine features for all sorts of reasons. And when you combine, if you need to do that, delta gamma coupling seems to be sufficient. And it seems to be a very efficient way of doing that.

Now, if you want to construct an unbounded array of, you know, hierarchically structured expressions, as I mentioned earlier, that might not be sufficient.

And so the empirical argument I'm making here is that delta gamma coupling will not be sufficient for the kind of rich syntactic phenomena that are often documented in standard textbooks, for example.

So anything beyond a basic phrase unit, I suppose, anything with long distance relations or dependencies and things like that, or in fact, anything beyond a single phrasal structure will probably need to be called upon all these other mechanisms that I mentioned earlier.

The more kind of richly layered levels of low frequency interactions.

So delta gamma is fine for, you know, feature combinatorics, but it doesn't seem to be sufficient.

Again, that's an empirical claim could be wrong about that, but it doesn't seem to be sufficient based on the literature for the more kind of complex structures of the kind that we that we all know in the linguistics world.

OK, and then we can also make some interesting claims here that I've made with my colleague Antonio Benitez Braco in in Seville in Spain.

And all these aspects of oscillatory dynamics obviously result from somewhere.

They don't come from nowhere. They result from genetic guidance.

And some recent workers provided a list of candidate genes for this particular guidance.

So we have a particular phase code, a neural code for syntax, but it's got to come from somewhere.

Unless you're, you know, well, I'm not going to name any names, but unless you're the kind of person who's so in love with, you know, deep learning and large language models that you think that it's all just some kind of epiphanominal domain general learning process.

But if you don't believe that, then there's got to be some some kind of genetic structure for language.

And so we've mapped a bunch of these potential genes using a bunch of different techniques and to particular particular functions.

But that's a very much an ongoing kind of far off project.

It's just the very beginnings of trying to map out the kind of genetics of the neural code for language.

But at least it's a coherent project, right?

It's a legitimate coherent project.

So, although rather nascent to, to be fair.

OK, so just to background this discussion a little bit more, I realize we're already quite far in.

So I'll just kind of skip ahead a little bit.

And one of the most important questions in the neurobiology of syntax is essentially, you know, what type of neural activity and low frequency activity, high frequency activity, spike rate can track very long or very brief phrases?

So my basic answer, which I've already essentially given to you, is associating roughly the respective scales of recording and neural organization with particular levels of linguistic complexity.

So low frequency gets you structures, high frequency gets you operations and spike rate individual spike trains give you basic representations.

OK, so this section here is slightly less important for our purposes, so I'll just skip over it.

But essentially, my claim in this section of the paper, section three, is just to kind of review existing models of language in the brain, neurobiology of syntax in particular, and just to show that they essentially tick many of these boxes, but not necessarily all of them.

So certain models emphasize the R or the O, but they don't talk about the S or the E.

Or in fact, in my own book, I talk about the O and the S, but I don't really talk about R or E at all, in fact.

Other work talk about potentially a combination of all these features, but don't provide a particular mechanistic relationship between how these features relate.

So, for example, you can talk about the neurobiology of R and O and S and E, but even if you have that, you still need to have some way of causally relating them, which is the kind of final section of this paper, which is

how you get.

So, for example, you know, I mentioned that the representations, the R unit is hosted at the spike level and the E unit encoding is represented in low frequency traveling waves.

But how the hell do you connect those two things?

Right. How do you directly what mechanism do you coordinate them?

So we know there's some recent work showing that, in fact, traveling waves can, in fact, directly coordinate spike LFP coupling.

That's that's some very promising work that to kind of unify all these features together.

But you've got to have a mechanism to do that.

So you've got to have a separate mechanism in between R and O, O and S and S and E.

It doesn't just come for free.

OK, so a lot of these previous models are very promising from a cartographic perspective.

They really do help us understand which parts of the brain are involved in different aspects of syntax or semantic.

But in terms of providing the how question, right, which neural mechanisms exactly are involved, that's typically lacking.

The best kind of that you have is definitely the work of Stan Dehane.

He's done some really nice work showing that signature syntax might be in places that most people have not really thought about.

Especially at the lower order level in individual spike, spiking activity or cellular barcodes and so on.

That's very promising and it's something that I invoke in this paper, in fact, as well.

OK.

So just to introduce us a little bit further.

These four components of rows, R or C, enter into every syntactic phenomenon.

Everything.

As soon as you get a syntactic structure, you need to have at least all of these things.

OK.

So in terms of representations, I define them just like any old mental representation, essentially.

And they are an instruction to provide a particular conceptual feature.

And they are composed of features determining constraints on operations as well.

And that's a really important point, by the way.

I do want to stress that.

It's well known in linguistics that, you know, words have things like selectional restrictions.

So a verb will often co-occur or will need to occur with a say in another word.

You can't just put words in any order that you like.

So, you know, words are kind of like, you know, magnets.

They're kind of trying to attach to certain other words.

And often that's independent of statistics, by the way.

There's some good work by Andrea Martin and Sophie Slat on the relationship between statistics of

language and structure in language.

And they're not the same thing at all.

They often correlate and they often aid each other in various ways, in very intricate ways.

But they're not the same thing at all.

And you can't simply derive all possible structures in language just from the frequency, you know, frequency properties of words and the bigram and trigram frequency rate.

You need something else.

You need kind of structural information.

So the fact, this is kind of maybe a subtle point, but the fact that individual representations do in fact have selectional and agreement and movement constraints already tells you that there's got to be some kind of relationship between or dependency between the nature of how we neurally represent these representations and how they then transition into the operation stage, right?

The O stage.

There's got to be an initial dependency.

So some of these features can be maybe a little bit thorough.

But, you know, linguists often talk about things like demonstrative, noun, plural, and so on.

And these are, of course, human defined.

You know, the way that our conceptual systems work, we've tried to define these things using these terms.

Are these cognitively plausible or neurobiologically plausible?

Maybe not.

But there's at least some kind of atomic unit that's being evolved here.

A unit, just call it X, right?

So this feature X and Y, they are being composed into particular configurations.

So maybe these terms, these English language terms, are just a convenience, a kind of placeholder for what these conceptual features really are, so to speak.

But I think that language is slightly misleading for a reason that people like Chomsky and David Popol have talked about.

A lot of these things really just come down to the kind of classic Mauryan levels.

You know, David Maury's three levels of computation, algorithm, and implementation.

All of these things are totally kosher and legitimate at the computational level.

And so therefore, it doesn't really, we don't really have to abandon them necessarily just because we can't find, you know, what looks like a noun feature in the brain.

Like, that's, you know, a lot of this involves kind of category errors, I think, so which is why I try and break it down into more atomic stages.

So syntax builds structure through recursive applications of merge.

Like I said, you have this basic merge operation, which I haven't introduced yet, I realize, but it's basically just a binary set formation with some kind of categorization.

So you form a set, you just put two things together, and then you categorize it.

You give a category, you know, a noun phrase or a verb phrase or something like that.

And there's much more to be said there, but that's essentially the basic property of language, just building sets, set formation.

Okay.

So this is the outline of the model.

At the R level, as I mentioned, you get single unit encoding of conceptual features and what are called formal syntactic features.

And these are basically features that I won't bother going into, but there are certain features like Q and Phi and other things that are, other features that are interpretable or uninterpretable.

Things that are features that only enter into syntactic relationships.

And don't have any of a role.

They don't have any of a role in providing instruction to phonology or anything else.

But so these features involve what's called a cellular barcode for distinct features that compose into syntactic objects coherently bound by this high gamma activity at O.

So the O stage is the later stage.

And it also involves vector codes.

This is invoking a lot of work from John Hopkins University and people like Stan Dehane again.

The R level also involves vector codes for ensembles, neural ensembles, cellular ensembles, hosting features common to objects represented at O,

which are ultimately coordinated by the S level, the low frequency level.

At the O level, high gamma activity.

So this high gamma activity is, like I said, it's a neural activity that is around 60 to 150, 200 hertz amplitude in the brain.

These high gamma sensory motor transformations.

So things like high gamma activity provides access to sensory motor transformations.

And they transform these sensory motor representations like sight and sound and orthography and so on.

Things that you read, things that you are exposed to.

They transform them into lexicalized objects.

And if you're interested in the particular regional involvement, there's a list there.

And then they are therefore accessible, as I mentioned earlier, they're accessible to delta or theta phase locking.

So I mentioned the theta gamma coupling, where the phase of theta dictates the amplitude and the firing rate of the high frequency oscillations.

So, you know, you only get that once you have high gamma activity, because that's the sufficiently, you know, kind of correct resolution.

The optimal resolution, it seems, which the brain likes to coordinate long distance relationships and provide, you know, transfer information from one part of the brain to the other, essentially.

So this level can implement the semantic composition of language specific concepts that coordinate the firing of our units.

So this high gamma level has also been implicated in basic semantic composition.

So in a paper of mine, I've looked at the composition of basic adjective noun structures using intracranial recordings.

And high gamma activity seems to be sensitive to this basic aspect of semantic composition, not necessarily syntactic composition, but certainly semantic composition.

It could also be involved in syntactic composition, but certainly semantic composition.

Syntactic composition, again, would necessarily involve the S level, you know, S for structure.

Okay.

And then, of course, the high gamma activity itself also activates these ensembles that I mentioned at the R level that are composed of distinct units hosting this barcode or vector code for the finite list of units that compose into the feature bundles that you get whenever you access an individual word.

Every given word is literally just a linear list of these features that you kind of trigger whenever you access it.

And again, you know, all of this stuff is only talking about the syntax semantic side.

The side of phonology and sound and gesture and orthography and all the rest of it.

That's a separate issue.

That's a totally separate issue.

I'm only talking about interpretation, comprehension, you know, planning, thinking, that kind of thing.

I'm just talking about internalization here.

Okay.

Turning to the S level.

As I mentioned, this is a low frequency neural program for generating structural inferences over the O level.

So delta theta phase ampute coupling, as I mentioned, gives you categorical inferences that allow you to modulate the representations of these feature bundles.

So the delta can coordinate the theta gamma complex.

And it has some say in the sequencing and the triggering and the kind of, like I said, yeah, basically the triggering of these features and how they are accessed.

Basically the readout.

What's the term, the readout, right?

And then finally, E, the encoding phase, this is what I mentioned earlier with the traveling waves, where you get the local and global workspaces for bottom-up lexical memory and top-down hierarchical memory.

Traveling waves implement delta theta coupling for hierarchical memory, whereas the theta gamma coupling gives you the kind of bottom-up lexical memory.

It's kind of linearized memory just for lexical features.

Whereas the hierarchical memory, the memory about hierarchical relationships in a structure.

So, you know, abstract relations in a tree structure, independent of any linear order or independent of any sentry motor transformations triggered by gamma.

The kind of separability between the structural inferences made and the lower level sentry motor operations that you get and are accessed by gamma.

And then as I mentioned earlier as well, alpha power can be involved in the shielding of this process.

An increase in alpha power towards the end of sentences is often indicative of, like, some kind of, you know,

accrual of semantic attention or semantic working memory or something like that.

So I think that's potentially related to this issue too.

And then finally, beta power also codes for syntactic predictions.

There's been some interesting work looking at how beta power seems to be sensitive to successful or unsuccessful, but basically triggered or not triggered syntactic predictions, or at least anticipation of building a syntactic phrase.

Or you're about to, you know, you've been exposed to certain words and now you're about to generate a structural inference or you're predicting a specific syntactic item.

Beta power seems to be involved in the indexing of that sensitivity.

But notice something interesting here, right?

That's beta power.

This whole notion of prediction and anticipation, that is a separate process from the core structure building inferential process involved in, that's been exhibited by delta and theta.

And has already been documented in delta and theta.

So again, this is a kind of neurobiological support for the separation of kind of, you know, predictive algorithms, predictive processing, and, you know, more kind of old fashioned traditional structural inferences that are made by the brain.

So, you know, just to unpack that a little bit more, it would be a problem for me, it would be a problem for a lot of people, if all of these processes, anticipation, prediction, you know, coordination, and attention, and also syntactic structural inferences, were all indexed and all coordinated in this delta and or theta rhythm.

That would be a problem because how the hell do you separate them out, right?

It's very difficult to think about that.

You know, if that were to be true, then we would probably have to abandon the notion that we can construct a coherent oscillatory phase code for language.

It might be impossible.

You just have to give up on it and look elsewhere.

But the fact that we can, in fact, find the separability, I think, speaks to this potential, which is why I wrote this paper, because I think, you know, I think there's some good potential here.

Okay.

And then this figure down here, if I can maybe just zoom in a little bit, potentially, this just depicts exactly what I mentioned in verbal form, in visual form.

So the representation level, again, these individual features here correspond to individual cells, sensitive, maybe not selective, but sensitive to each of these features.

The question of selectivity versus sensitivity, that's purely empirical.

I enjoy, you know, philosophy and theorizing as much as anybody else, but these questions are, at this stage, purely empirical.

It's possible that there are certain cells that are selective or, you know, sensitive to all these features, but at least that's the framework.

There's a particular cellular infrastructure that facilitates these representational accessing, essentially.

And when it comes to operation, again, I need to be careful with the term operation, because I, maybe this is

the one that I haven't explained well enough, so I should maybe spend a minute just explaining this.

There's, the brain, you know, carries out phonological operations.

It carries out semantic operations.

It carries out syntactic operations.

It carries out all sorts of operations.

In the context of rows, I want to, I'm focusing on how it carries out the combination, the initial set formation process involved in merge.

So the early stage of merge, essentially.

The O versus S kind of captures the distinction between merge and labeling, essentially.

The distinction between set formation, just building, you know, putting two units together, and then transferring it into a category.

So I guess what I'm trying to say is the categorization process will not be found at the O level, at the high gamma level.

You might see some category sensitivity in high gamma.

You might see different gamma profiles for, you know, verb phrases versus noun phrases.

And in fact, we've already found that.

People like Andrea Mara has published papers on this.

So we know that there's a syntactic or semantic sensitivity at the O level.

Again, the difference between verb phrase and noun phrase is not just syntactic.

It's also semantic, right?

That's the problem.

Big issue here of trying to separate syntax from semantics is that whenever you modulate syntax, you also modulate semantics, and whenever you modulate lexico semantic content, you also, therefore, modify everything else, syntax and semantics.

So that's an issue.

And there's, you know, we're trying to, well, we are, in fact, you know, running some designs at the moment to try and experimentally tease those apart.

But for this purpose is, for the current purposes,

the term operation is referred to unifying different lexical and semantic features together, essentially, into a coherent unit, into a set.

And then once you've got that set represented in gamma, then you turn to the structural level.

And the structural level is more higher order, more complex, more global, more interactive, and therefore more low frequency, right?

That's the kind of goal.

That's the kind of framework here.

And then finally, the encoding level.

I mentioned very briefly, maybe a little bit too briefly,

the two types of memories that you get in language.

There's been some really good work by people like David Adger, Queen Mary, and a bunch of other people who've kind of explored the different types of syntactic workspaces or linguistic workspaces.

Like when you comprehend a sentence, you don't just have a single, well, you might, in fact, have a single workspace.

It's possible.

But it's also possible that you read it out into different separable workspaces that are responsible or sensitive to different types of memory representations.

So it's often been implied by people like William Machen,

I think very legitimately,

that inferior frontal cortex is really involved in syntactic working memory.

Specifically, we don't know exactly what features of syntax,

but certain types of syntactic information are held there

and maintained there in short-term memory.

But so for lexical memory, I'm invoking theta-gamma coupling.

And then, as I mentioned briefly, for the hierarchical memory,

I'm involved, I'm invoking delta-theta coupling.

And then unifying the both of them or connecting the two

was this other form of processing that I mentioned earlier,

the delta-gamma coupling, right?

So the delta-gamma coupling, as I mentioned,

is just involved in basic domain general,

very domain general, combinatorics,

just combining two different representations

from different parts of the mind.

But nevertheless, you'll see the relationship between delta and gamma

between both of them.

Now, when it comes to the spatiotemporal dynamics of this process,

the actual kind of, you know,

how this is implemented in the brain,

in terms of which networks,

which white matter pathways, and so on,

that's a purely empirical question.

But we do know enough from the MEG,

from the scalp literature, from the intracranial literature,

that these codes are legitimate codes to be implemented

across some portions of cortex.

I have a bunch of speculations empirically supported
also in terms of which parts of the brain
these particular encoding workspaces are going to be found
and going to be documented.

But that's at least,
this is the general abstract schema, if you like.

This is the kind of very general schema.

And the reason why I've emphasized that
is because, as I mentioned earlier,
every other paper in neurolinguistics
focusing on syntax is almost entirely cartographic.

It's almost entirely localizationist
in terms of let's figure out if it's, you know,
is it IFG?

Is it PT?

You know, is it ventral temporal?

They're the kind of questions that are framed
by most of these models in the literature.

And though essential,
that's not my concern here, right?

My concern here is,
yes, localization,
but let's try and focus on
which particular abstract neural code
is going to be generalizable enough
across different parts of the brain.

And also, by the way,
small point here,
also generalizable across cases of plasticity.

You know, I'm involved in a neurosatria case today.

We know that the brain is highly plastic.

When you resect certain portions of tissue
recruited or not in language,
you get fairly rapid organization
over a period of weeks and months,
sometimes years.

But that tells you immediately.

And in fact,

there's also really fascinating work
in congenitally blind people
and or deafblind, I think, too,
showing that portions of their,
yeah, in the congenitally blind,
there's portions of occipital cortex
that come online
and are sensitive to syntax.

So because, you know,
these people are not using
their occipital cortex for vision,
they don't need it,
they end up recruiting it
for other purposes too,
sometimes linguistic purposes.

So that's why I think focus
on a more abstract code
is more important.

Maybe not more important.

Maybe that's unfair.

But I think it's potentially
just as important,
if not more important,
than trying to figure out
the kind of cartographic landscape
for how language is implemented.

Also, by the way,
another small point,
which is maybe not so small these days.

This is also very important
if we care about things like
brain-computer interface devices.

So, you know, neurotechnologies.

So there's a bunch of
what's called BCI,
brain-computer interface devices,
that you implant in patients' brains
to help them

with either their epilepsy
or to help them
with speech processing.
You get quadriplegic patients
who lose their speech.
And there's some other efforts
recently to try and help people
with all sorts of language deficits
by using neural implants.
There's a factory
not too far away from me,
north in Austin.
And there's places,
you know, owned by Elon Musk,
as you know,
who owns Neuralink.
These guys are trying
to do something similar.
But, you know, that's for speech.
When it comes to using,
designing a brain-computer interface
for things like syntax
or semantics,
I think we're going to need
to pay attention to these aspects,
these questions,
to do with the neural code
rather than just the region.
Because if you're going to have
somebody who has a syntactic deficit
or a semantic deficit,
and you're going to try
and help them
get their language faculty back,
it's no good just knowing
which part of the brain
is going to be recruited
for their language function.

With a BCI device,
you need to coordinate
particular stimulation profiles
and particular ways
to coordinate and trigger
and suppress brain activity.
And so in order to do that,
you've got to have
the right parameter space.
And to figure out
the parameter space,
you therefore need to have
a question of neural code involved.
almost definitely a phase code,
some kind of high-privacy
or low-privacy phase code.
So all of these questions
that I'm asking right now
don't just have
kind of theoretical,
you know,
basic science academic implications.
Obviously, that's my interest.
That's my main concern here.
But they will also definitely have
kind of, you know,
implications for patient treatment
in the BCI space
moving forward.
Okay.
Right then.
Let's move on.
Let's skip past some of this.
And I've...
It's getting on to
almost an hour now.
So I'll skip through.
The next few sections

are just outlining
representation,
operation, structure,
and encoding.
So I'll just pick out
some of the most
important moments from here.
But again,
this section number five
just defines,
redefines, again,
and maybe exhaustively,
what I mean by representation,
some example features,
and in fact,
what these gamma oscillations
and high frequency oscillations
might be indexing more generally
in terms of their cognitive scope
and their computational kind of facility.
This is just a little brief
definition of the kind of scales
that are relevant
to the different levels.
So we have single unit activity,
multi-unit activity,
the envelope of the spike,
and the LFP.
These are different aspects
of neural organization
and recording resolution
that will be relevant
to the different components
of rows.
And that kind of just outlines
the scope of inquiry here.
This was something
that I put in to satisfy,

make it clear that,
you know,
I'm talking about representation
in this sense
and not that sense.

Basic units,
again,
I'm just talking about
what these basic units are.
and then I'm turning here
to concerns about
how to distinguish it
from things like operations.

Okay.

And then,
oh yeah,
so this is an important paragraph,
I think.

So this recent work
by Nelson F. L.
discovered preferential activity
for content words
over function words
in the anterior temporal world.
This is single unit recordings.

Again,

I should stress,
sorry.

So that already begins
to tell you that
at the single unit level,
at the silo level,
different parts of the brain,
I mentioned ATL already.

In this case,
ATL does seem to be sensitive
to lexical type.

Again,

that could be due
to other reasons.

In this case,
they looked at it
independent of word length
and word position.

The big confound here
is word frequency,

but,

you know,

there's some other kind
of indication here

that potentially
something to do

with lexicality

is driving this difference,

this sensitivity.

Okay.

And then more generally,

what are termed

concept cells

that preferentially

activate for stimuli

that occupy

a certain position

in some category axis

seem to support

a declarative,

seem to support

declarative information.

What does that mean?

It just means that

there's some really nice work

by Bauer et al

showing that

there's a kind of

feature space

in a primate

and also human
ventral temporal cortex
that is sensitive
to a kind of
axis of animacy.
So whether or not
an object is
animate or inanimate,
like a chicken
versus a laptop,
there's some things
that are kind of
in between and vague
and then whether or not
a surface is spiky
or stubby.
So kind of,
you know,
like this or spiky.
So kind of,
kind of a kiki booba
kind of situation
for linguists.
So that's kind of
a nice conceptual
geometry that you
have going on.
And there's different
cells,
concept cells
that preferentially
activate the stimuli
that occupy a space
within that axis.
So that tells you
very nicely
that at the unit level,
we have a potential

neural code
for this
very,
you know,
generic,
but also all-encompassing
object space.
This is Doris Sauer's lab,
the Barakow paper.
So,
you know,
the kind of
promise,
the promise that this
gives us
for linguistic features,
I think is quite
interesting because
linguists have spent
a long time,
you know,
reminding us that
the feature space
in linguistic semantics
can also occupy
something like
animate,
inanimate,
and then some other,
a bunch of other
abstract concrete
kind of feature spaces too.
So I think it's very
reasonable to,
to expect
and to predict
at the unit level
a similar kind of,

you know,
conceptual geometry
taking place
in a different part
of the brain,
potentially posterior
temporal cortex,
potentially inferior
frontal cortex,
you know,
who knows?
We don't know
until we explore,
but I think
that kind of,
you know,
reinforces
the definition
of the R level
that I'm defining here.
Okay.
Right there.
Okay,
so this,
this next section here,
mapping from R to O,
this gets at the,
the very tricky issue
that I mentioned
at the beginning.
So at the beginning
of the paper,
I,
you know,
lampoon and criticize
all these other people
for not doing this.
So hopefully I do it,

you know,
a decent job,
otherwise I will
embarrass myself
a little bit.

But the kind of
motivation here
is that we don't
just need a definition
of what R and O
looks like in the brain.

We need a way
to relate R to O.
We need a way to feed
the information
at the R level,
the unit level,
all the way up
to the O level.

So I mentioned
just a quick,
a quick reminder,
R is a single unit,
you know,
spike train activity.

The O level
is high gamma activity.

Right?

So they're kind of
slightly more
easy to detect
using standard
intracranial measures,
for example.

Okay,
so how do we relate
these ideas,
the O to the R?

So in this section,
I kind of propose
that the relations
between multi-unit activity
and population signals
hold the answer.

Okay?

So I might just
have to focus on this
for a little bit.

Oh,
this is only a short section.
Very short section.

That's good.

Okay,
this will be quick then.
So local field potentials
reflect the common
synaptic activity
of a population
of neighboring neurons
and spikes are
short-timed
high-frequency
content signals
reflecting individual
cellular activity.

Right?

So individual cells.

Neural synchronization
can be evinced.

One of my favorite words there.

They can be evinced
by temporally relating
spiking activity
to the background
oscillations of LFPs.

And this relationship

has been documented
across multiple brain regions
and cognitive functions.

So this spike phase coupling
has functional consequences
and there's some examples
if you want to consult.

Okay,
so that's the coupling
between the LFP
and the actual spike,
the individual spike.

So that's kind of
an analogous, right?

If you think about it,
that's analogous
to what I mentioned earlier
in the low-frequency space, right?

I mentioned the delta-theta coupling
and the, you know,
the delta-gamma coupling,
et cetera.

This LFP phase coupling
is essentially the same thing,
just on a smaller scale, right?

So it's kind of nesting
all the way down.

So LFPs are highly
effective means
they're exposing
what state
a given cortical region is in
since they capture
general dynamics
not specific
to any individual cell.

And this is a really
cool point, actually,

about LFPs.

They actually host,
you know,
types of information
that are not detected
at the spike level.

So that's a very,
kind of an important point there.

And also, of course,
you know,
there are cells
that don't spike too, right?

So in the same way
that there's likely
much information available
at the LFP level

that's not represented
at the unit level,
like dynamics
and amazing properties

exist that can only be
detected at the level
of summed activity
of a million neurons,

so too is it expected
to be the case

that there are aspects
of cortical computation
that are only represented
at the interactional
global scale,

even at the state scale,
like, you know,

hidden Markov scale,
kind of,

autoregressive,
hidden Markov models,
and so on.

You can kind of generate
these more abstract state spaces
across networks of brains,
networks of brains,
I'm sorry.

And I think that will also
be relevant at some point
too in the future,
but maybe I'll get to that
at some point later.

And that information
is not represented
at the LFP or spike level.

So that basic finding,
empirical finding,
is another core motivation
for the Rose model, right?

The fact that you do
find this
dissociable level
of information representation
across these scales
already tells you,
it already tells you
that there's very likely
to be this scale
of computational complexity
at the linguistic
and non-linguistic level too.
Okay.

And like I said,
it's a basic presupposition
of Rose,
but it's also a presupposition
of other major frameworks
in cognitive neuroscience,
right?

Not just me.

I'm not the one

to kind of,

you know,

propose this.

So an example

I'll kind of hear

is working memory,

which seems to involve

discontinuous bouts

of spiking activity

as opposed to

steady state neural dynamics,

right?

So often it was initially

believed that working memory,

in order to maintain

something in working memory,

there's got to be

some kind of direct,

intuitively direct

neural correlate,

which looks like

it's maintaining

or staying at a stable level

or, you know,

amplitude,

steady state

or increase over time,

et cetera.

Turns out not to be the case.

Okay.

So relating R to O

is far from trivial.

Yes,

that is absolutely correct.

Consider how communication

through coherence

has typically been

assumed to reflect
phase synchronization
between oscillators.

Recent work has offered
an alternative mechanism
through which coherence
is the consequence
of communication
and emerges
because spiking activity
in ascending area
causes postsynaptic
potentials in the same
but also other areas.

That makes things
more complicated.

And so these authors
identified
afferent synaptic inputs
rather than spiking
entrainment

as the principal
determinant of coherence,
opening up new directions
for framing the relation
between units and coherence.

So LFP coherence
appears to be determined
by two factors.

Coherence due to the
direct contribution
of afferent synaptic inputs
and coherence
between the sender LFP
and the sum population
spiking activity
in the receiver.

So therefore,

coherence therefore
depends on connectivity
strength
and oscillation power
and does not need
purely oscillatory coupling
or spike phase locking
in a receiver.

And then favor of
complexities arise here.

I mentioned some examples
from our neighboring field too
but I'll kind of skip over that.

It's an interesting topic.

But still,

you know,

successfully relating
the two fundamental signals
in the section,

spike and LFP,

can provide us

with a comprehensive

explanation regarding

the neurobiological cognition

and since many signals

picked up by the LFP

will also very likely

be able to be found

at the unit level,

care must be taken

to map out

assembly level effects

from single unit responses

and that's kind of

an empirical portion.

So this kind of section

is very,

I think it's very honest,

I think it's very upfront
about the limitations
and potential,
you know,
countervailing trends
in the literature
that provide some obstacles
to this grounding.

But nevertheless,
I think even if these things,
you know,
are true,
there's still the possibility
for multiple types
of phase codes,
multiple types of neural codes
within the same space
being involved
in different types
of cognitive operations
or across different
core knowledge systems.

And of course,
you know,
given that we know
that human syntax
is in fact
a potentially species
defining property,
I think it's,
you know,
care must be taken
to really separate out
the,
separate it out
from these constellation
of domain general
neural mechanisms

which might be relevant
but might also
not be relevant,
right?
It's again,
that's,
you have to kind of
weigh up the likelihood
of a domain general
versus non-domain general
process being relevant
for your concern.

And again,
you don't know
until you know,
you don't know
until you do
the empirical research.

But I think,
you know,
the purpose of this
subsection
is to kind of
open up the space
of,
you know,
possible alternative
mechanisms.

Okay,
I'm going to skip
over this section.
This is just discusses
gain field mechanisms.
I've written a few
papers about this.
It's a very interesting
topic,
just kind of grounding

the kind of lower level
accounts of why,
you know,
what it is,
what it is we know
already about
what the computational
properties of
individual cells
might in fact be.

And that might sound
a bit abstract,
but there's some
really good work
by W.

Decombe-Sofitch
and Randy Gallistel
and a bunch of
other people
that I cite here
just exploring
the potential
computational
kind of,
you know,
kind of classical
Turing-level
architecture,
you know,
Chomsky hierarchy
kind of computational
facility
that these individual
cells might have
in terms of
their operational
power.

Okay.

And then this subsection here just introduces a kind of a more classical distinction between Sherringtonian and Hopfieldian views. That's basically just a distinction that focuses on the transformation of signals by nodes in a point-of-app architecture, viewing cognition as the result of patterns of nodes and node connections, as opposed to viewing representational spaces through which computation is considered to be the transformation between spaces. So it's kind of a more, different ways of viewing populations and units, the relationship between units and their population dynamics. And I kind of

speculate that,
you know,
evolutionary older
brain areas might
be explained better
via Sherringtonian
accounts,
whereas more
recently evolved
structures,
hint, hint,
language,
right,
might not be.

So that's a
potential avenue
that could also
direct
a way of
exploring the
kind of empirical
space beyond this.

And then,
when it comes to
the operation space,
I've already mentioned
much of this stuff.

I'm invoking
high gamma activity.

I am talking
about the
different,
oh yeah,
well,
there's different
dimensions again,
right?

Gamma is not

just a unified
construct,
either neurochemically
or genetically.
In fact,
it's a very complex
structure indeed
across species,
across brain regions.
You get physiological
gamma responses
in subcortical structures.
You get kind of
pyriformocortical
responses in cortex.
Gamma itself
is just,
you know,
literally it's a range.
But it's a very
complex manifold.
So I kind of,
this section
kind of takes care
to break down
what I mean by gamma.
You know,
a lot of papers
in the Neurobiology
of Language
just kind of invoke,
you know,
gamma oscillations
or particular
arbitrarily cut off,
you know,
range,
band range.

This section
tries to take
a little bit of care
to just unpack
the different types
of gamma,
different types
of aspects
of gamma
in terms of
their sensitivity
to different aspects
of representational
complexity.

So lower
versus high gamma,
you know,
broad band,
narrow band gamma,
etc.

If you're interested
in that,
this is a good section
for you to consult.

Okay.

So turning to structure,
this again,
kind of reviews
some of the things
I mentioned earlier
to do with the fact
that, you know,
every linguistic structure
is sensitive
to some aspect
of lexical
and semantic information
that cohere

into some kind
of phrasal
or sentence structure.

There's a phrase
by Alcimorans,
there's no escape
from syntax.

That's absolutely true.

You know,
everywhere you look
in language,
every kind of phenomenon,
semantic,
pragmatic,
even phonological,
there's often
a very direct impact
of syntactic information
in terms of
the coordination
and the instructions
and the influence
it has on these
other levels.

Syntactic structure
really is the most
kind of fundamental
level of language
that influences
a bunch of different
domains.

I've already introduced
merge,
so I'll just kind
of skip over this.

This is just a kind
of more set theoretic
classical definition

of what we mean
by a feature combination
of the kind involved
in language
and of the kind involved
in the kind of work
that I mentioned earlier
and of the kind involved
in the hyperfrequency
gamma and theta dynamics
that I mentioned
earlier too.

So merge is basically
the most fundamental
operation in language.

It's the basic property
of language
as Chomsky calls it.

Okay.

Again,
from a more
philosophical perspective,
I always found it
interesting that
even though this is
a very rudimentary
set formation operation,
if you think about
the cognitive consequences
as people like Peter Haglott
have in a very nice paper,
the computational system
of language
seems to be related
to neural signals
integrating perception
and action
and providing humans

with novel modes
of planning
and interpretation
whereby lexical units
and unification processes
like merge
provide an imaginary space
that transcends
the influence
of direct perception
action cycles.

That's a really key point
there because of course
we know that
one of the
real benefits
of language
and potentially
why it was
evolutionarily selected
is because it allows
us to plan,
interpret,
think,
consider personal
responsibilities,
consider our own
future,
our past
and so on,
construct a notion
of one's past
and where you've been,
construct a narrative
so to speak.

So we know
that other animals
have memories

but maybe
none of them
have a past.
Maybe only humans
have a past.
It's kind of a
unique kind of concept
that involves
some kind of
a different
epistemological transition
and a linguistic
transition too,
right?
It's a fundamentally
linguistic notion.
Same with the self,
right?
The concept of self,
you know,
who I am
and who you are,
I,
are you and so on.
The existence
of the pronoun system
in language,
for example,
all of these things
are extremely
cognitively rich
and semantically
very important concepts.
They're not just there
to facilitate
communication.
They're not just there
to facilitate

information transmission.

They really are part
of the mental
architecture
that we have
as human beings.

So I think
it's an important
point that Huggard makes
the fact that
this really is the
center of the kind
of cognitive
neurosciences.

Okay.

Oh, excuse me.

This section
just introduces
syntactic features,
what we mean by
particular syntactic
feature.

I mentioned
lexical features
quite a lot,
but this section
kind of just briefly
touches on
syntactic specific
features.

And it also
reviews some
recent work
looking at
the role of
endogenous oscillations
in neural computation.

I mentioned

some of the stuff
earlier to do
with low frequency,
sensitivity,
and entrainment
to particular
moments of
syntactic structure
building.

This section
here just kind of
unpacks some of
that more exhaustively
and gives some
more recent
references there.

So this section
kind of just reviews
that literature.

Skipping through
this part here.

And again,
just kind of
to recapitulate
what I mentioned
earlier,
the role of
lower frequencies
in indexing
supralexical
structural inferences
is very well
empirically supported.

So the previous
two pages that I
just scrolled past,
they really kind of
propose,

summarize a lot
of literature
suggesting that
lower frequencies
indexing supralexical,
so stuff above
the individual
weird level,
it really is
pretty undeniable
at this stage,
I think.

Okay.

And then we
turn to the
more issues of
timing.

So, you know,
what is the
precise timing of
syntactic
information processing?

This paragraph
here just kind of
breaks down
some of the
particular periods
of sensitivity
that have been
already documented
and how it is
compatible and
maps onto
the predicted
time course
dictated by the
high frequency
rhythms, right?

So it's no good
me just saying,
oh, delta is
involved in
syntax and,
you know,
gamma and,
you know,
semantics or
whatever.

You really need
to have a
motivated time
course in real
time, like a
real time passing
constraint on how
these high frequency
oscillations can
embed themselves
and be entrained
and phase locked
to these lower
frequency rhythms.

So I kind of
just break down
the time course
of activity here.

And it's all
kosher, it's all,
you know,
empirically motivated,
it's all kind of
aligned with what
we know about both
about delta and
about the
intermediary

frequencies too.

Okay.

And then this section turns to, again, the mapping process that I mentioned, the mapping from the O to S structure.

So this kind of invokes, again, a separable mechanism.

I'll just scroll up to the top here just to have a quick refresher.

So we had phase amplitude coupling, as I mentioned, right?

So this is spike phase coupling mapping R to O.

Mapping O to S is phase amplitude coupling.

So that's the kind of classical process that I mentioned earlier to do with theta gamma and delta gamma coupling because that gets you directly to the higher order structural inference.

Okay.

So this section
is just a short
review of the
phase amplitude
coupling dynamics
in this particular
process.
and also talks a
little bit about
some other kind
of extracranial
recording measurements
that have kind of
unveiled this in
other kinds of
domains.

This section here,
dissociating structure
from meaning, this is
very much for the
linguists.

This is kind of
trying to figure out
how we can carefully
account for signals
that are specific to
the semantics of a
phrase versus
signals that are
maybe not specific
but certainly sensitive
to the syntax of a
phrase.

That's a very tricky
issue.

You have to kind of
basically design an

experiment or stimuli
that is getting
participants to pay
attention to
structural inferences
and syntactic
information and
syntactic acceptability
and not pay too much
attention to other
kinds of features.

But also the stimuli
itself have to be
carefully weighted
across words that
preferentially engage
syntactic information
over and above
conceptual information
or conceptual semantic
information versus
words that very much
do trigger heavy
semantic content.

So that involves a
lot of careful
kind of empirical
consideration.

Our lab is currently
carrying out some
research into that
space as are some
other labs too.

But it's basically
an appeal for
focusing more laser
like on these
ways to dissociate

syntactic from
syntactic from
semantic processing
using empirical
novel kind of
experimental paradigms
because a lot of
the paradigms that
are used at the
moment have a big
problem which is
they don't really
carefully separate out
moments of syntax
from moments of
semantics.

So it's very
difficult to kind of
realistically
and kind of
confidently make it
clear that a given
brain signal or a
given response is
really being driven
by either syntax
or semantics.

Okay.

In terms of
skip over this?

No I didn't.

Okay.

And then this
outlines the
mapping from the
S to E levels
and it talks a
little bit more

about some of
Standerhane and
other people's work
into vector codes
population dynamics
to do with what
level of
representation we
might find
individual syntactic
features at.

And again I
transition slowly
here to talking
about traveling
waves.

How the kind of
the more kind of
static code that I've
presented so far
in terms of just
you know a delta,
a theta, a gamma,
how that static code
might be
transferred into
something more
you know cognitively
and neurochemically
plausible in terms of
the what we know
about the
transformation of
information across
white matter tracks
and across how
different portions of
cortex, cortical

cortical information
is kind of
transferred in real
time across
different portions of
the cortical mantle.
It's not just like
stationary.

Okay.

So I guess what I'm
kind of calling for
here and as I call
for in the next
section is a more
kind of spatio

temporally
dynamic model
or code for
language essentially.

Excuse me.

So all of this
section essentially
needs, it kind of
calls upon the
notion that you
need to map on
an important
you know important
findings from the
psycholinguistic
domain in terms of
the timing of
language, the kind
of you know when
we know that
language is coming
involved, which

periods of activity
are going to be
important and the
actual neural
architecture.

That's kind of a
very tricky
question.

How you map on
those.

So for example you
know if a
psycholinguist says
okay we know that
when people using
eye tracking or
using scalp

ERP responses, we
know that the
period of 300 to
400 milliseconds is
when most people
kind of detect
semantic violation
effects.

We really then have
to say okay if
that's the case does
that time period at
the behavioral level
map on to moments
of you know neural
activity also at that
level or maybe
potentially the neural
response comes on a
little bit earlier
before it's detected

at the scalp level
or maybe at the
single unit level
it's even earlier
or vice versa or
maybe there's a
kind of top-down
effect whereby some
kind of intermediary
mesoscale neural
signal can be
detected first before
the spiking
information is.

You know there's
different hypotheses
that arise here in
terms of the
directionality and
causal relation
between the R to
E levels.

In this in the
Rose paper I make it
you know very clear
what my empirical
predictions are but
you know other
alternative ways to
modify the Rose
model could be in
terms of the
directionality from
different levels right
so for different
linguistic structures
it could be the
case that some of

the more mesoscale
levels in fact
influence the
coordination of
spiking rather than
spiking information
being read out at
kind of high order
levels which then
facilitate the
inference is being
made.

The directionality is
very clear in the
paper but it's kind
of open it's very
much open for
empirical you know
resolution essentially.

Okay so in this
figure, figure two,
figure one is the
basic kind of you
know components of
the model.

Figure two is the
basic mechanisms of
the model.

So this just goes
into some more
technical details that

I won't go into to
do with the actual
nature of spike
LFP coupling what
the kind of
mathematical
foundations of it

are and how you
can you know map
that on to a kind
of classical kind
of low frequency
response and how
these different
signals could be
coordinated in real
time.

Okay.

And then turning to
the encoding section
as I mentioned
earlier you know
we know that the
human mind needs
different workspaces
for different
aspects of
cognition.

So the big
question is does
language share a
workspace with other
kind of you know
non-linguistic
processes?

Is there a
specialized unique
workspace for
linguistic
information?

it's possible.

It's possible.

I think it's likely
in fact which is
why I have it in

my paper.

So in fact we
know that there
are some non-human
primates that can
execute very
primitive combination
processes like
potentially even more
thoughtful composition
just combining two
calls you know like
primary vocal
calls together into
a basic primitive
unit.

Can they then do it
again and again and
again?

Can they do they have
the workspace that
allows the facilitation
of storing this one
unit and then
creating a new
structure and then
you know assigning
that whole complex
an identity that is
independent from its
discrete parts which is
what merge is all
about in language.
they don't seem to be
able to do that.

So that tells you I
think quite reliably
that we do need some

kind of notion of
human specific
syntactic workspace
which is what the
whole encoding level
is meant to be in
rows.

Okay so this section
kind of just plays
out a little toy
example of what how
to derive a
particular structure
and what the whole
process would look
like going from R
to O to E.

So I think this is
important for me to
kind of outline
since it's the part
of the paper where
I kind of really do
give a concrete
example.

So let's consider
the sentence
old men walk
slowly.

During the
comprehension of
the first two
words the delta
gamma combinatorial
code coordinates the
feature bundling of
the atomic data
structures hosted by

old men.

So old men have

various features.

At a minimum this

involves posterior

superior temporal

sulcus, low

fixed activity coupled

with neighboring

posterior temporal

cortex but also

cross cortical sites

responsible for the

specific feature

types in question.

So I've already

mentioned places

like ATL and

IFG.

And then I also

mentioned theta

gamma coupling.

This maintains in

short term memory,

I mentioned the

lexical memory

buffer before,

theta gamma coupling

maintains in short

term memory the

relevant units via

high-through

activity in a

linear sequence.

So this is

worked by David

Popel too.

The theta gamma

coupling is a linear feature clocking mechanism that tells you these are the set of features, features 1 to 5, 1 to 7, and you're going to access them and trigger them and read them out in a given sequence.

So that's a kind of linear lexical memory stuff.

It doesn't give you the hierarchical syntactic relational component that is independent of order that comes later.

Okay.

At the transition between men and walk, so word 2 and word 3, the superordinate delta theta code maintains the categorical identity of the object.
object.

So, you know, old men is obviously men that are old, right?

As I mentioned, it's not an old quality that happens to be men.

So that's a noun
phrase.

When you get to
old men walk,
you relabel the
object and it's
no longer a noun
phrase, right?

You have to
reategorize the
syntax.

It's not a verb
phrase.

It's old men
doing something.

It's an event
structure.

So things like
event structures and
quantification structures
and all the rest of
it, they involve
very specific, unique
semantics that are
informed and
provided by and
configured by the
syntactic category.

So that's why I
mentioned that, you
know, there's no
escape from syntax,
right?

The syntactic category
feeds the semantic
information.

Okay, so when you get
to the third word

walk, the
superordinate delta
theta code maintains
the categorical identity
of the object.

In this case, the
negotiation between a
multi-unit noun
phrase and a more
complex verb phrase
hosting old men.

So during the same
period, the lexical
memory code, again, if
you remember the
lexical memory code
is the theta gamma,
the lexical memory
code increases its
number of theta
nested chunks due to
the occurrence of
walk, right?

So old men walk is,
in terms of the
theta gamma code,
all the theta gamma
code sees is another
word.

When you get to
walk, all the theta
gamma code sees is a
third word.

That's it.

It just sees, okay,
another word's coming,
another word, another
word.

And it just simply
chunks those features
together using the
feature clocking
mechanism that I
mentioned.

But that's not what
the delta complex
sees, right?

That sees something
else.

So the superordinate
delta theta code
maintains the
identity, and in
this negotiation
between, and in
this case, the
negotiation between a
multi-unit noun
phrase and a more
complex verb phrase
right?

During the same
period, the initial
lexical memory code
increases the number
of feet and nested
trunks due to the
gans' walk.

The same transition
occurs from walk and
slowly, with the
exception that while the
lexical memory code
still increases in
coupling strength, and
that's the measurement

of pack relations
between the feature and
gamma, right?
High pack relations.
The hierarchical memory
code would decrease
closer to, but not
identical, its pre-verb
baseline due to the
adjunction relation not
demanding a revision of
the hierarchical memory
representation.
So what the hell does
that mean?
So when you get to
old men walk, you
change the syntactic
category.
It's a verb phrase.
So that involves an
activity at the delta
theta level.
When you get old men
walk slowly, the word
slowly does not change
the syntax.
It's still a verb
phrase.
So old men walk
slowly is a verb
phrase.
Old men walk is also
a verb phrase.
So slowly changes the
lexical semantic
information.
So you keep getting

that theta, gamma
pack increase.

But you get either a
steady state or a
decrease.

Again, I predict the
decrease of the delta
theta pack relationship
because all you're
doing is maintaining
the same syntactic
identity.

You don't need to
change the syntax.

Same syntax.

Again, I talked about
syntax feeding semantics.

If you think about it
for a second, old men
walk slowly is still
involves old men.

It could be old men
walk quickly, old men
walk, you know,
lethargically, whatever.

It's still the same
thing.

So the fact that old men
walk slowly doesn't
change the fact that it's
still old men.

Whereas with old men
walk, that does change
the fact about old men.

So when we get to old
men, we have to say in
fact, we know about old
men.

When we get to old men
walk, that changes
things.

We know that it's old
men walking.

When we get to old men
walk slowly, that's a
modification structure.

It's called an adjunct.

The adjunct structure or
adverb or prepositional
phrase, anything like
that, simply, it
adjoins.

So that only involves
basic set formation.

It doesn't involve a
relabeling of the
category itself because
the category is still
the same.

So this gets into some
theoretical syntax ideas
that are not important
here.

But the basic idea is
that you just get an
enhancement of lexical
semantic information,
but the syntax is
identical.

So that's why you get
this kind of wave of
old men walk and then
slowly.

And this is, you know,
I mentioned some of the
delta peaks before.

This kind of stimuli,
old men walk slowly,
is what has allowed
researchers to figure
out that you get those
low frequency peaks at
the third word.

Old men walk.
slowly.

Because that's the
period where you
really change the
syntax.

And it turns it into,
you know, a kind of
sentence, I guess, a
kind of full sentential
structure.

With a propositional
identity, you know,
truth evaluability.

It can be true, it can
be false.

You can afford all sorts
of different epistemic
judgments to it.

That's the whole kind
of, you know, you kind
of get a whole complex
of instructions being
sent to conceptual
systems at that stage.

Okay.

So the transfer of
relevant lexical
information from
categorization or
labeling that I

mentioned would take
place via interactions
between these two
neural codes, right?
So separable codes.

I speculate here
potentially via theta-theta
phase-phase coupling or
phase locking of the
theta-gamma workspace and
the delta-theta workspace.

Theta-driven dynamics
effectively constitute the
handoff of information
after lexicality has been
established by lower-level
R and O processes,
transitioning from encoding
lexical memory to multi-object
memory.

Direct testing of these
dynamics specifically with
respect to syntactic
workspace construction has
currently not been
undertaken, although much
work has been carried out
demonstrating increased
theta-gamma coupling in
human hippocampus during
memory formation, as well
as enhanced frontal theta
to posterior gamma
coupling, alongside the
recent discovery of a
rapid neocortical theta
network mechanism for
flexible information

encoding.

And familiar operations
from left-corner
mentalistic grammars can be
appropriately parselated onto
these R, O, and S levels.

That's a bunch of jargon,
but the linguists will
appreciate what I'm talking
about there, I hope.

But future work should still
treat this issue of precise
parsing models with more
care and tact than I have
demonstrated here in this
initial architectural
proposal.

So what I'm trying to say
there is that I mentioned
Ma's computational,
implementational, and
algorithmic level in the
middle.

I've talked a lot about
computation and
implementation.

I've only spoken a little
about the algorithm, the
thing connecting the wet
wear, the kind of
biological stuff you can
touch and see, with the
kind of more abstract
algorithm in between the
computation.

And that's the domain of
what's called
psycholinguistics.

So the field of psycholinguistics has a bunch of different parsing architectures for how humans parse sentences.

There's rival models.

Lots of them have different predictions for how human beings parse different types of weird sentences in real time.

I have my own favorite candidate.

I've mentioned left corner minimalist parses.

That's kind of my own favorite account for all sorts of empirical and theoretical reasons.

But it's an open question.

And in fact, later I kind of discuss the possibility that different parts of the brain might implement slightly different parsing models.

So, you know, the posterior temporal cortex might be a good place to find the middle misgramma.

But the inferior parietal cortex or the inferior frontal gyrus might be an excellent place to find a

different kind of
parsing model.

And I cite a bunch of
examples in the paper
somewhere in here, I
think.

But that's, again, an
critical question.

But it's also kind of a
more, I think it's a more
pluralistic way of kind of
viewing language in the
brain.

You know, it's not just
like, you know, minimalist
parsing grammars explain
everything.

And the entire brain is,
you know, in concordance
with the predictions of
minimalist grammars.

It might be a kind of
more, a different kind of
constellation of parses
that we need.

But still, nevertheless,
with the basic primitive,
you know, computational
architecture being
inflected by moments of
merge and labelling and
kind of more traditional
basic assumptions from
general grammar.

Okay.

This moment just, yeah,
this period just turns a
little bit more towards

traveling waves.

And a little bit more to
do with fMRI research,
kind of building the
paper more to what we
already know from the
extracranial world.

This section talks about
syntactic memory as being
phase synchronization over
successive cycles.

So this invokes a bunch of
interesting work and
consults some models from
the world of
mineral biology of
oscillations more generally
in terms of phase dynamics
and cross-trips
coupling and all the rest of
it, just to kind of
situate what the model
might look like in a more
kind of mathematically
rigorous feature space.

And the section on
symbolic computation
turns a little bit more to
what we already know
about, in fact, I think I
should review this very
quickly because this again
comes back to the issue of
domain general study.

So, region directions and
the neurobiology of
navigation and memory are
relevant to the conception

of E.

So, hippocampal, cortical,
sequence replay and
encoding is not constrained
to simply repeat past
experience.

Rather, this process is
informed by an internal
model of the world,
generating representations of
inferred entities, not
necessarily encountered
physically.

Again, this is one of the
reasons why I work with
people like Carl Friston,
because I think this whole
notion of building a
general model of one's
environment, the interplay of
action and perception
cycles and so on.

That's really a fundamental
negotiating building block
of what language gives us,
of what the role of
language really is in the
human mind.

It's really about
interplaying these two
different divides of the
human brain.

So, this active generative
capacity motivates the
authors to propose that
replay in the brain
instantiates a form of
compositional computation.

So, a given replay sequence constitutes a set of entities strung together into a compound, whereby each entity is bound to a representation of its compound role, determining its function as part of a whole.

This establishes a clear separation, with respect to composability, between entity and role, or what's called syntax and semantics, right?

The entity and the role.

So, while roles encoded by hippocampal cortical interactions can certainly be spatial, they can also be non-spatial, and even non-spatial and non-Euclidean, potentially involving arbitrary roles, such as a verb, or what we would call verb, right?

Some kind of event anchor.

So, the entity role bindings currently explored empirically in humans are limited to things like which position and which sequence, but if other roles like if, then, else can be encoded in a similar way, then replay may form a viable candidate for a

neurophysiological
mechanism implementing
symbolic computation.
So, this compositional
nature of replay is
implemented via our
friend theta-gamma
coupling, mirroring close to
the present assumptions of
lexical feature sequencing
and basic semantic
compositionality being
complemented via the same
dynamics and high
gamma activity.

And there's a nice paper
recently by Nina Kazanina
and David Popel that
explored very similar
notions.

They kind of argue that a
lot of the neural
mechanisms already known to
be found in hippocampus
could be used for a
symbolic language of
thought in language.

Language thought meaning
the kind of classical
Fedorian notion.

So, you know, a lot of
what's already known
outside of language about
the neurobiological
infrastructure for things
like navigation could be
utilized and exploited by
the language system in

order to achieve its goals
of composable functions,
compositionality, and
basic structure
building.

So, that's, again, once
again, it's an empirical
question.

These are all empirical
questions.

But I think the, you know,
the purpose of this paper,
which is a purely
theoretical paper, is to
kind of negotiate through
this terrain and see which
candidates are more
feasible, more likely,
et cetera.

And I think this candidate
in particular, I'm not
the only one who thinks
this, is a very, you
know, viable candidate
for grounding some of
the initial, again, the
stresses, for me it's
initial.

It's just that initial
phase of, the early
phase of merge, the
basic set formation
process, the theta
gamma stage, potentially
not the, you know,
human-specific
structure building stuff
that I mentioned

earlier to do with the structural inferences that involve other mechanisms that I mentioned.

But at least in some aspects of language, lexical search, lexical access, you know, morphologically complex word binding and so on, it's likely that these things are potentially used as well.

Okay.

And then in terms of, so memory transfer, this is an important concept that's been raised recently by working by Brinkett at AI and Elle Miller.

This is just, this section just very briefly reviews how we might transfer memories across the cortex, which kind of might look a little bit like transferring syntactic structures into a workspace or transferring stored lexical items into a separate kind of consolidation or monitoring workspace.

There's some kind of potential for research in this domain.

And then turning to alpha-beta dynamics, this is an interesting topic that I mentioned briefly earlier to do with how the, you know, the role of these oscillations that are outside the main code space that I already mentioned, right?

I've already mentioned, I've delimited my code space.

What about the role of alpha and beta?

I've mentioned it briefly, but it's kind of quite an important role given that they are quite dominant.

They're very prominently found in intracranial and extracranial research into language.

So just focusing on syntactic memory, there's a recent paper, Garrick et al, which supports a role for beta in syntactic identity, just being sensitive to the identity of a sentence, the type of

category that you
afford it.

These authors
investigated speech-
memory representations
using intracranial
recordings in the left
paracelabian cortex
during delayed
sentence reproduction
in patients undergoing
awake tumor surgery.
Based on the memory
performance of patients,
they found that the
phase of prototemporal
beta represents sentence
identity in working
memory.

The notion of
sentential identity
presupposes a labeled
or categorized structure,
like a verb phrase,
seemingly represented
partially by
frontotemporal beta.

And converging with
other literature,
beta may represent
aspects of the global
cognitive set going
beyond syntax-specific
information to include
conceptual and
statistical information.

Again, I mentioned the
role of beta in

syntactic anticipation,
syntactic prediction,
the kind of statistics
of language.

And I think that has an
important role to play
here.

again.

Intercranial recordings
by yours truly of
auditory language
comprehension also
implicate frontal alpha
and beta power in
phrase prediction and
anticipation.

They kind of mark the
moment of potential
anticipation of a
licensed phrase.

Another paper found
that both beta and
gamma are sensitive to
syllable rate, but only
beta power is modulated
by comprehension rates.

So comprehension
obviously implies, you
know, semantics and
syntax rather than just
syllables.

And over the past
decade, in fact, a
little bit longer than
that, over the past
15 years or so, there's
been a debate about
whether beta power

effects during sentence
comprehension reflect
syntactic computations
or instead reflect
maintenance or set
updating.

And to briefly
summarize the most
current results, it
appears that the
latter maintenance
hypothesis is most
well-supported.

So I'm therefore
going to put further
discussion of this to
one side.

But the basic role of
it in maintenance in
terms of just
continuing, aiding,
with the assisting in
some way of
maintaining the set of
generated and sustained
cognitive representations,
not necessarily being
sensitive to the
properties of this set,
not being sensitive to
the particular syntactic
identity or the
category of the syntax
or even the category of
the semantics or the
features of the
semantics, but at least
being involved in

coordinating neural
resources, available
neural resources, to help
with the sustaining of
that memory in a
current space.

Again, this is very
domain general, not
specific to language,
which is why I think
you find fewer of the
language-specific syntax,
semantic sensitivities.

Even though it's
involving comprehension,
as I mentioned,
comprehension means
everything.

It's just like, you know,
processing a sentence
means loads of things.

It doesn't mean
specifically syntax,
specifically language-specific
stuff.

It just means, you
know, when you process
a sentence, you do all
sorts of things.

So I think it's most
likely that it's involved
in these kind of more
domain general maintenance
processes.

And then in parietal
cortex, alpha enhancement
seems to index syntactic
working memory demands.

Much like the regulation of gamma by alpha in control and attention mechanisms that I mentioned earlier, parietal cortex may play an important role in memory and complexity, such that low-frequency rhythms originating in lateral or medial parietal cortex regulate the activity of gamma encoded operations in lateral temporal cortex.

And single unit or assembly encoded are in medial temporal or inferior frontal cortex.

That's a more kind of specific proposal.

And another likely site of syntactic working memory is in period frontal sulcus for reasons cited in this paragraph.

It's very heavily involved in all sorts of interesting semantic integration processes.

And then, yes, as as I mentioned, bringing this model to a kind of conclusion, these assumptions take place all within the context of the travelling

wave framework.

So lower frequencies migrating across the cortex, particularly being relevant to cases of maintenance and storage.

So traditional standing waves lead to periods when all neurons in the network are turned off, whereas travelling waves can ensure that sub-portions of a network remain consistently active, directly compatible with the mosaic-like architecture of posterior temporal cortex in semantic integration.

I won't explain that further, but it's basically the idea that within these small regions of cortex, like posterior temporal cortex, there's a very interesting patchwork of activity of little satellites of sensitivity to different aspects of language.

Even though they're all generally sensitive to language, within them there's kind of more

specialised regions,
which is not unlike what
you find in the
temporal temporal cortex for
things like face and
place and all the kinds
of shape perception,
kind of ventral visual
occipital cortex being
having a kind of
tapestry of, you know,
cellular, cell-specific
sensitivity to different
representational features.

So where exactly the
waves travelled to during
phrase composition is,
again, my favourite
phrase.

This is purely an
empirical question, but
some candidate regions
have been suggested
about phrase.

That's kind of where we
stand at the moment.

That's where the field
is at the very moment.

So figure three here is
the basic processes of
the ROWS model.

This kind of just gives
a general outline of
how ROWS is
hypothesised to be
implemented across the
frontotemporal language
network.

And at the bottom,
there's a representation
of the various
frequency interactions
proposed to implement
phrase-fixy building.

So you have
representations,
category-specific
information being
encoded in spikes.

With operations, you
have broadband gamma
activity.

Initially, I got
hypothesised in PSTS
based on the
intracranial work that
we've done, but it
could be anywhere.

It could be in PN2G,
but also be in portions
of IFG, depending on
the category potentially
of the phrase, the type
of structure you're
generating.

That's an open question,
very much open.

In terms of structure,
we have the low-frequency
neural program driven by
temporal cortex.

And then for encoding,
we have these travelling
waves that spatially
migrate structure.

So this, I also think

that, given what we know about the speed of travelling waves, I think it's possible that a lot of this stuff transfers up either a door, either a door sort of ventral pathways across posterior to anterior temporal cortex and from posterior to anterior frontal cortex, given the time scales of sensitivity that we know exists there. Okay, this final section of the paper talks about causal evidence, which is increasingly becoming very important these days in neuroscience as we've exhaustively mapped correlational activity of loads of different linguistic processes, but we haven't really got a good sense of which parts of the brain are actually essential for language. So when you get regions X, Y, and Z being activated, you know, region X might be causally involved, whereas regions Y, and Z might just be activated or, you know, recruited or implicated, but not actually essential for the function.

So this section just kind of charts out some of the ways of exploring that empirically through cortical stimulation mapping in the OR or in the AMU, using TMS, for example, or joint TMS MEG studies, that's a possibility.

There's all sorts of ways you can think of kind of providing different portions of causal evidence for these models.

And these paragraphs here just kind of just review mostly non-linguistic, but also some language research into how people have used these methods to disrupt language in particular parts of the brain.

And there's some more interesting research disrupting particular phase codes and trying to particularly trying to modulate either hypo or hyperexcite and particular oscillatory dynamics or pack dynamics in different parts of the brain.

Now that's a really good way to directly test the causal involvement of a phase code more generally because if you can hyperexcite a theta rhythm

or hyperexcite a gamma rhythm, then that can really screw up your potential, your facilitation of the coordination of these electrical features and your attention and working memory and all the rest of it.

So that's a great way to kind of causally test not just a brain region's involvement, but also a particular brain, like how a brain region is involved.

Okay.

This section is much more philosophical.

If you guys play Elden Ring, you might get the reference in this section.

But the point of this section is to kind of evaluate the Rhoes model, to kind of situate it in a large context, to relate it to existing models in the intracranial space and extracranial space too.

So some of the important periods in this paper.

So I've already mentioned this paragraph, I think, to do with how different parts of the brain might be sensitive to different parts, so I'll skip that

bit.

This paragraph here just outlines how the very basic philosophy in Rhoes is that mesoscale neural organization can be useful for brain function and that high frequency gamma oscillations report mechanisms and underlying communication channels of neural computation.

So neural oscillations organize cortical activity to produce computation and Rhoes builds an architecture for syntax that presupposes this.

Some exciting prospects for testing Rhoes come in the form of multi-channel recordings with broad cortical access using binar microelectrode arrays, etc.

And I just cite some more technical examples there for people in the field.

And then, let's see, what else do we have here?

Yeah, I've already mentioned the inside-out outside it.

Well, actually, no, I haven't mentioned that.

I've mentioned Mars framework.

But just very briefly, I

think my model is
incompatible with a bunch
of recent ideas to do
with encouraging people to
explore the brain from
inside out.

So I'll just read this
paragraph and then maybe
try and digest it a little
bit because it's probably
the most important
philosophical kind of
grounding here.

So brain models informed
by computational concerns
will continue to be
needed in neurosciences.

Yes, that is true.

In particular, as we
approach the advent of
widespread availability of
single unit recordings.

And again, one of the
anxieties I have in this
paper is that as, you
know, linguists get their
hands on sophisticated
neural recording measures,
we really need to be
careful and quite
conservative in knowing
what we can say with
each of these measures.

So if you give every
linguist on the planet
access to single unit
recordings and the human
brain, you're going to

get a lot of papers
making claims about, as I
satirize here, neo-
Davidsonian existential
closing neurons or, you
know, specific, very
niche kind of cellular
types responsible for all
sorts of Baroque and
complicated and not very
cognitively plausible
linguistic constructs.
So I really wanted to, in
this paper, boil it down,
you know, decompose language
into its primitive
components, decompose at the
computational level and
decompose at the
representational level.
That kind of goes back to
the 1980s kind of classical
framework of the
computational
representational theory of
mind, where you have a
series of fixed
representations, units, and
fixed computations or
things that you can do
with those units.
There's kind of actions and
objects, right?
That's a basic metaphysical
human distinction between
things and processes,
actions, objects, and all
the rest of it, right?

Nouns and verbs, and
computations, representations.

So I really wanted to make
it clear that, you know,
next time you find
yourself with access to a
single unit, you know,
data set, you need to kind
of have sufficiently
decomposed these levels to
make it kind of neuro
chemically and neuro
biologically plausible,
rather than talking about
the kind of more high
falutin stuff that I
mentioned earlier.

So it's kind of more,
it's more of a cautionary
tale, I guess, because I,
you know, other fields have
gone through this before,
right?

So for example, you know,
grid cells, right?

It's over the last decade,
grid cells in interinal
and paracampal cortex were
initially implicated in
spatial navigation, then it
turns out they're involved in
auditory navigation, and
then they're involved in
conceptual navigation, a
certain of words, they
ended up just being
involved in navigation,
full stop, just all types

of navigation.

And I think a similar kind of transition might happen with when we get an increasing number of publications about language in single units, you'll get a lot of people initially saying, oh, well, you know, these units, these specific cell types are involved in X, then it turns out they're also involved in Y and Z, and we'll ultimately end up being forced to propose a more generic account for these things, which is why in my paper, I try and, you know, focus it, I try and appeal to as much generosity as I can.

There's always going to be some domain specificity, because obviously language has domain specific representations, and apparently domain specific computations, so we will need some notion of domain specificity on some level, but the only question is like, where are you going to find that?

Okay, I'm rambling a little bit, so I'll just, I'll continue.

So this outside-in perspective has been critiqued recently.

Juju

Brzecki

advocates for an insider perspective on building neural models of cognition, and considers the classical Maurean framework for purely outside-in perspective.

Yet Ma himself stressed that the three levels should be investigated in parallel, not necessarily prioritizing any given level.

So this is a balancing act though, right?

Too much outside-in, and you're going to get these neo-Davidsonian neurons, too much inside-out, and you're going to be told that linguistics departments need to shut down, right?

So regardless of your own philosophical bent, I think single unit and other types of intracranial recordings are plainly the most direct and reliable means to further test and refine rows, in particular given recent independent assessments and

critiques of bold
activity.

But, you know, the basic
message here is that a
purely insider perspective
is when people will say,
well, what's called dated
driven science or data
driven neuroscience,
right?

Where you say, well, let's
just do a bunch of
sophisticated statistical
analyses on our single
unit data and see what
kind of emerges.

Let's just hope that
language will just arise
somehow without any
ideological presuppositions
or any kind of biases.

First of all, it's
impossible, I think, for
any human being to do
that, even if they tell
themselves they're doing
it.

You're always going to be
delivering some biases.

You're always going to be
bringing some philosophy of
science or philosophy of
mind to the table, whether
or not you like it or not.

On the other hand, you
have the completely
outside-in perspective,
which I'm also very much

against.

So the outside-in
perspective would say,
let's, you know,
everything that linguists
say is Bible.

All these theories of
double-weight movement
and pipe-piping and all
the rest of it, they are
literally what the brain
cares about.

And we have to find that
in the brain.

We have to keep our fixed
theories of
higher-order cognition
and just look at the
brain and expect the brain
to care about everything
we care about.

We expect the brain to be
sensitive to all the stuff
that we as linguists care
about.

And the problem with
science is that, you know,
there's always a negotiation
going on.

There's always a kind of deal
that needs to be made.

So, you know, one of our
former presidents talked
about the art of the deal.

In this case, the art of the
deal in this case is figuring
out how much linguistics you
need to hold on to and stay

true to and how much you need
to let go of and how much
you need to say the brain
doesn't care about this.

You know, it only cares
about this, but not that.

My own, so my own feeling
is that the brain, excuse
me, the brain does genuinely
care about some aspects of
language, domain-specific
or otherwise, namely the
representational feature
types and also this business
of merge and labeling that
I mentioned, the basic
computations.

I think the brain does
genuinely care about that.

It probably doesn't care
about all the other stuff
that linguists talk about,
but it can be characterized
at some kind of more
epiphenomenal level, I guess,
as some kind of emerging
phenomenon.

We can talk about the
philosophical details, you
know, in other kind of
frameworks, if you like,
but that's the kind of, I
think, way to go about.

I think it's much too harsh
and arrogant to just
completely abandon all of
the insights of contemporary
linguistics just because we

have sophisticated
neuroscience.

I think some, holding on
to some insights from
linguistics will be
essential to direct and
test and, you know, refine
our data-driven science
too.

So I think what Brzezaki
talks about in his book
is, I think, misleading
and kind of a, you know,
unhelpful division.

I think it's not, I don't
like to talk about
inside out or outside in.

I think it's kind of just
you have a balancing act
between both, right?

It's, you have to just
negotiate when it's
appropriate to use
computational levels and
when it's appropriate to
use implementation level
theories.

Okay.

So that takes care of
that.

I'll skip over this bit
here.

Yes.

So this, this, this gets
into what I mentioned.

I mentioned this maybe a
little bit briefly, but it's
very common to object to

these kinds of proposals of
the kind I mentioned in
rows as a kind of crude
form of direct mapping,
right?

Or isomorphism.

And I saw some people
listening to me right now
might think, well, you're
kind of a hypocrite because
you're doing exactly what
you're critiquing Brzezaki
were doing, right?

I'm complaining that
Brzezaki is just, I'm
sorry, I'm complaining
about these outside in
people are just imposing
their linguistic theory
on the brain.

And I'm saying you
shouldn't do that.

At the same time,
though, the whole point
of the rows model is to
directly map on what
different types of
linguistic structure onto
the brain.

So a complete nihilist
or, you know, might say,
well, the brain really
doesn't care about any of
these structures.

The brain really doesn't
care about atomic
features, lexical items,
feature bundles, phrases,

you know, sets of sets
of sets and all the rest
of it.

Now that might indeed
be true.

That's totally possible.

I think it's unlikely
given that the, you
know, the specific
responses that we've
seen at different levels
of linguistic complexity,
the kind of sociable
scales of sensitivity
that you can see at
different neurocomplexity
scales that I've
mentioned here.

But so, you know, the
rows model is essentially
invoking a direct mapping
in some way, a kind of
one-to-one correspondence
between a scale of
linguistic complexity
X and a neural
mechanism Y.

And you kind of, you
know, you follow that
all the way up until you
get to the highest level
of nesting that you get
with the delta-theta
complex, all the way
down to the spike at
like peak coupling that I
mentioned earlier,
right?

Okay.

And so some people
might say that's a
problem.

Rose continues this
crude tradition,
aha, and by
establishing an even
more explicit isomorphism
between system-level
complexity and
syntactic complexity.

So, for example,
cells and syntactic
features versus global
coherence and phrase
structure on the other
hand.

Alternative accounts that
build syntactic
operations into neural
systems of diverging
levels of structural
complexity have not been
forthcoming, by the
way.

So if they do
forthcome, then we
can test them as
well.

But we'll also
naturally be required to
establish any kind of
adversarial collaborations
and experimental testing
of the models.

So through this, it
would be possible to

pit different candidate
neural mechanisms for
syntax against one
another and compare
effect sizes across
studies to see which
mechanism, either from
Rose or some other
model, best explains
real-time direct
vertical effects of
language processing.

So that's a nice way to
directly test the Rose
model.

But I also want to
emphasize here that,
as far as I know,
the Rose model is the
only model of syntactic
syntax, non-neurobiological
model of syntax, that
takes seriously this
whole notion of the
fact that the brain
genuinely exhibits
distinct levels of
neural complexity
organization and
scales of complexity,
and also takes
seriously the fact that
not all neural
recording measures just
measure the same
thing, right?

Single units up to
Utahs, up to

SEGs, up to
ECoG, up to
MEG, up to
SCALPEG.

These are very
different beasts and
they record very
different things.

It's easy and
convenient and
maybe comforting
just to say, well,
the kind of signal
that we see in
SCALPE RP responses
and delta waves are
probably just some
kind of summated
phase reset of
whatever's going on
at the spike level.

On some level of
description, that's
true.

However, it doesn't
help with building a
model of syntax in
the brain.

That's just kind of
dismissal.

So my appeal here is
to kind of ground it
more concretely in
different levels of
analysis.

And so far, you
know, all the other
models of syntax in

the brain, as I've
said, are
localizationist as
opposed to kind of
algorithmic.

Or that's why the
subtitle of this
paper is a
neurocomputational
architecture because
it's a neurocomputational
architecture and it
is very much an
architecture.

There's within the
Rose architecture, you
can imagine three or
four or five, you
know, very different
specific theories of
or models from the
architecture.

So, you know, it's
very much an
architecture.

And within this
paper, I propose my
own specific theories
at each level.

But the scale of the
architecture scaling from
the spike of the
coupling up to
pack up to
traveling waves, that's
the architecture.

But so within the
architecture, you could

maybe reframe some of these things as long as the mechanistic relations between R to O to S to E are maintained.

That's the kind of general framework here. R to E.

And then going back to metaphysics for a moment, Rose is also sympathetic to recent moves in philosophy of biology to view a range of biological constructs as processes rather than objects.

Rose is built entirely from neurocomputational mechanisms that are already known to subserve clusters of generic, perceptual, and cognitive operations.

I'm not invoking any new neural mechanisms.

I'm not proposing a new kind of microtubule quantum computational mechanism to explain all this.

I'm being very, very conservative.

I'm just using literally stuff that's been published in the most high impact, serious

scientific journals,
which has been afforded
textbook level treatments
at every level.

So all of these
mechanisms have been
given serious textbook
treatment.

The only difference here
is that I'm migrating
some of these
concerns over to the
linguistic domain with
various different
motivations across the
levels.

Okay.

And then, turning
slightly further, so
yeah, another nice
point here made in the
Nature Neuro paper by
Nina Casanina and
Alessandro Tabano.

Current theories that
model hierarchical
structure building via
low frequency dynamics
correlate some neural
measure with attributes
of a hierarchical
syntactic structure and
thus concern the
outcome of syntactic
structure construction.

So this is a critique of
those MEG and EEG
studies that show this

low frequency delta
entrainment or peak
response at moments of
syntax.

It's a very legitimate
critique.

But by initiating
syntactic structure
building at the cellular
R level and
accounting for how
these output low
frequency responses at
the S and E levels,
the RhoES architecture
goes beyond these other
accounts that are more
closely tied to the
output of structure
building.

So, and again, even in
the RhoES architecture,
this is correct that
these low frequency
delta peaks are indeed
the output of the
structure building.

I'm not denying that.

I'm not saying they're
the input.

You know, they're not
some kind of
intermediate response.

Casanina and Tabana
are completely correct.

They do reflect some
kind of readout, the
final stage of the

syntactic structure
building process, some
kind of wrap up or
even monitoring stage.

For me, it's part of
the structured to
encoding phase.

But at least the
early stages of syntax
of the R and O
levels are definitely
at this kind of spike
level and high gamma
activity level.

Okay.

Yes.

And then again, open
questions remain about
how to ground this in
more neurochemically
explicit models.

I haven't talked much
about neurochemistry.

Some of my earlier
papers in 2015, 2016
tried to take a
neurochemical framework
much more seriously
and I have more of a
mathematically rigorous
framework into
negotiating particular
low frequency dynamics.

But in this paper, I'm
just kind of leaning
that to one side for
now because the
paper is already, you

know, way too long.

And I'm just focusing
on the kind of
architectural components.

That's something to be
focused on at a
later stage.

Okay.

Yes.

Yes.

Yes.

Okay.

So turning to the
conclusion.

You know, I've
mentioned a few times
that the Rose
architecture really
does come in
response to the
absence of phase
coding models in
language.

There's been a lot
of phase coding
responses or phase
related, you know,
entrainment and low
frequency responses
empirically documented.

They've been
empirically documented,
but in terms of the
actual theoretical,
you know, summary,
that's kind of been
lacking.

So that's my main

motivation.

And many researchers
who have kept the
frameworks emerging
purely from
extraneous, you know,
meg derived event
related fields or
scalp EEG related
potentials or
fMRI responses,
et cetera.

These guys have
begun to lose
hope.

They've begun to
lose hope in the
prospect of finding
syntax in the brain.

So for example,
Lena Pilkanen,
who's probably,
you know, the
world's most
respected neurobiologist
of syntax.

She's written that
the neuroscience of
language field has
long assumed that
our brains build
syntactic structure
during language
processing.

Today, it's
reasonable to
question this
assumption.

And she's, you
know, she's
totally right and
motivated to say
that because she's
summarizing research
from mostly
extracranial stuff
like MEG and
fMRI.

And she's
definitely right
that if you just
focus on that
research, it seems
pretty hopeless.

I think the lack
of spatial
temporal, the lack
of high spatial
temporal resolution
in these methods
explains the
absence.

That's kind of one
of the implications
of this paper.

And this paper
is essentially
indirectly critiquing
that field.

So Pilkenin, you
know, reasonably
speculates that
based on the
evidence reviewed
in her article,
something like

MEG might not be an obvious neural operation, or alternatively, it might not be found in presently explored neural signals.

In fact, she has this nice distinction between syntax as knowledge and semantics as process, where in, you know, neural recordings across all methodologies, finding signatures of semantics is very easy, because semantics is very energetically costly, and the number of words and possible concepts you can combine at any given moment is really kind of astonishing.

You know, the number of different feature spaces you can talk about, you can talk about green horses made from licorice that ride to Mars next Tuesday.

You can talk about
all sorts of
surreal stuff,
right?

So semantics is
very unpredictable,
measurable, and it
calls upon a wide
range of cross
cortical representational
features.

What about syntax?

Syntax is very
different, because
syntax is just
merge.

That's it.

You just build, you
just, you put two
things together, and
then you label it.

So that's it.

There's no
different, there's no
novelty there.

Every time you
build a phrase, you
do it in the same
way.

You build a phrase
architecture, you
label it, you
stack it into a
workspace, you
label that, you
ship it off to an
interpretation and
procedure, and

that's it.

That's all of
syntax.

And you do it
again and again
and again.

You don't have a
different merge
process on Tuesday
as you did on
Friday.

But semantics is,
like I said, there's
a big difference
here between the
space of semantics
and the space of
syntax.

The space of
syntax is just
this merge, this
one operation.

So it's no
surprise that it's
been very
difficult for
scientists to
find neural
signatures of
syntax, syntax
specific responses
because it's like
we do it so
reflexively and so
trivially and we've
been doing it, you
know, since infancy,
maybe even before

infancy, depending
on which language
acquisition theorist
you believe in.

Whereas semantics,
we're constantly
changing it.

You know, it takes
years and years and
years for us to
really build our
full repertoire of
developed semantics.

With syntax, once
you've built a
phrase, you've
built a phrase.

That's it.

You've got all of
syntax.

The only thing that
you can go beyond
that is, you know,
kind of performance
aspects, like how
complex a sentence
you can construct,
right?

Can you read
Shakespeare?

Can you read
Joyce?

You know, can you
read David Foster
Wallace?

That's a very
different question.

But that's a

question about
performance.

It's a question
about like working
memory, attention,
you know, all those
sorts of things.

When it comes, the
actual basic component
of syntax is just
merge.

So I think
Pitkenen's right
that, you know,
maybe she's not,
I don't think she's
right that, you
know, merge's
knowledge and
semantics's process.
I think they're both
process and they're
also both knowledge
too.

But in terms of why
it's been difficult
for people to find a
signal reliably,
it's because that's
all there is to
syntax.

Whereas with
semantics, you're
going to get a very
reliable, very
serious neural
energetic cost
involved in processing

sentences.

So, and in fact,
in our lab, in our
intracranial recordings
for epilepsy patients,
all the response,
all the most reliable
responses you see are
clearly with, you
know, meaningful
sentences as opposed
to, you know,
semantically impoverished
sentences that have a
good syntax.

Because it's just
difficult to really
disentangle those
things.

But that leads to
some other questions
that I won't get
into because that'll
take us to a
different topic.

But so, you know,
I guess I've tried to,
what I've tried to do
with this paper is
that with Rose, I
think I've at least
provided some
candidate neural
signals.

And they do exist.

There's plenty of
candidate neural
signals, some very

good ones.

And that can, you
can be readily
investigated in this
manner.

So, yeah, I think
that kind of sells
that issue, at least
for me.

So, under Rose,
there's basically no
sense to be made out
of claims that
syntax lies in one
specific brain region.

I talked about
PSTS before being
the real site of the
initial phase of
merge, but in order
to get to structural
inferences and then
transferring that to
workspace, that's a
global phenomenon.

It involves most
likely, you know,
posterior temporal
inferior frontal
coordination.

There's a nice
paper in PNAS by
Oscar Wolno, who's in
our lab, a postdoc in
our lab, who shows
very convincingly that
building a coherent
sentence structure

involves the
coordination of
overlapping,
spatiotemporally
overlapping, but
distinct clusters in
both posterior temporal
and inferior frontal
cortex.

cortex.

So, it's not just, you
know, IFG versus
PT.

It's kind of like,
okay, they're both
involved, but to what
extent?

Some issues are a bit
tricky to discuss
because you have to
kind of relate this to
the lesion literature,
which kind of more
reliably implicates
posterior temporal
cortex strongly.

And for my money, if I
were just to, you
know, contradict myself
just a little bit, I
think the language
region is essentially
posterior superior
temporal surface.

I think PSTS is
basically the language
region with other
things coming online

later due to either
coupling or functional
coupling and
coordination with PSTS.
PSTS will always be
involved.

It's always the hub.
And there's other
regions like IFG and
ATL are involved and
called upon for
performance reasons or
reasons to do with
downstage readout and
interpretation.

Well, PSTS is the
language region, for me
at least.

So that's kind of the
seat of the rows
localizationist approach
at least.

But then the rest of
the neural code spreads
far beyond that, as I
mentioned.

I only read one
sentence of that
paragraph before
waffling.

But I think I pretty
much really said that
in many ways.

Okay.

And then, yeah, we
can think about
testability in all
sorts of different

ways too, to do with
separate out the
category of the
phrase versus the
semantic of the phrase.

That's something to
think about.

Yeah.

Okay.

And then in terms of
the general roles in
shaping information
processing, the
currently identified
frequency-specific
mechanisms seem to
align with an
emerging consensus in
the field that slow
frequencies control
input sampling, alpha
and beta gate
information flow, and
high frequency
activity is controlled
by slower rhythms.

So that's a very
general kind of
neurocomputational
framework that's been
explored in the
code of neurosciences.

And then moving
beyond this, a newly
emerging focus on
brain criticality and
how this concept might
relate to brain

rhythms provides
exciting revenues for
future work in
theoretical neuro
linguistics.

So that's a cool
concept too.

And in fact, there's
some really good
work exploring how
in some of my
earlier papers, I
talked about what's
called globularity,
which is this
concept of the fact
that the human brain
is uniquely almost
spherical relative to
other primates, not
completely spherical
obviously, but a lot
of cognitive abnormalities
come with a brain
shape reorganization.

And the fact that the
human brain is kind of
more globular than our
ancestors has been
used to motivate a lot
by a lot of serious
anthropologists and also
neurobiologists, this
idea that the human brain
is kind of uniquely
efficiently wired in
some way to facilitate
cross-cortical and

also sub-cortical
cortical information
integration.

So I think there's
something to be said
there to do with, you
know, an evolutionary
change in brain shape,
globular brain shape,
would have naturally
facilitated different
paths for these
traveling waves to
move across, right?

Different patches of
cortex can communicate
with each other.

And in fact, we know
from work by people
like Liz Spelke that
the basic computational
kind of, you know,
contribution of
language is to provide
a kind of universal
currency or different
conceptual domains to
talk to each other.

Excuse me.

So a language is
basically giving you,
you know, you have
then your number sense,
your sense of
morality, your sense of
intuitive physics,
intuitive geometry,
intuitive botany,

what is all sorts of
different domains in
the psychology literature
concerning how human
beings have intuitive
modules for
understanding
reality and
constructing a
general model of
the world.

Language seems to
uniquely allow us to
conduct transactions
between these domains,
whereas in non-human
primates, they may not
be encapsulated exactly,
but they are much more
kind of atomized.

The modules, you know,
non-human primates have
most of these modules for
sure, no doubt, maybe all
of them, but they don't
have a way of conducting
transactions between them.

Whereas with humans, we can
talk about, you know, a
large number of, you
know, funny, green,
colorful, heavy balloons
or whatever, right?

We can combine different
conceptual categories
together.

And number sense,
morality, you know,

different geometrical
constructions, emotional
constructions.

We can use language to
access all of these
different cognitive
constructions.

Now, even more
interestingly, there's
some, you know, if you
look at the kind of
syntax of natural
language, there's been
some really interesting
work by people at
Queen Mary University
and other universities
showing that if you
take the list of human
semantic concepts and
the list of human kind
of, you know, electrical
items and features,
there's a lot of
overlap in terms of
like event structure
and agent processes
and features.

but there's also
some non-overlap.

So there's certain
concern, certain
notions like worry
or concern, for
example, that are not
morphologically marked
in any language,
whereas things like,

you know, trustfulness
or belief and, you
know, things like
that are marked by
words and language.

So there are certain
conceptual features
that are very readily
found across the world's
languages and have
certain morphological
inflections for or
features for.

But there are also
certain epistemic
notions or notions
pertaining to knowledge
and truth and so on
that are not readily
morphologically marked.

So that suggests that
there are certain parts
of the brain, certain
cortical modules that
the language system does
connect with and does
functionally speak to and
interact with.

But there are other
cognitive or perceptual
modules that it does not
speak to, that it's
more isolated from.

So that's kind of a
potential way to kind of
think about ways to
relate the kind of
neurobiological

infrastructure with the
kind of behavioral
linguistic kind of
phenomenon.

Okay.

So, yeah, and then just
to conclude, the more
kind of philosophical
foundations of ROSE are
also in line with
research documenting a
rich array of innate
capacities utilized during
language acquisition in
Neonics, which can
distinguish distinct
phonemes, detect word
boundaries, layered
words and so on.

I'm kind of really
invoking a very rich kind
of array of innate
capacities.

So currently, due to the
lack of consensus
regarding how to ground
syntactic combinatorics
and X-Kelty in the
brain, researchers enjoy
many degrees of freedom
when selecting from their
preferred linguistic
theory, processing
theory, the algorithm
and neurobiological
framework, the
implementational
framework.

So I hope that ROSE basically allows us to further constrain the landscape to narrow the space of likely candidate neural mechanisms for syntactic structure building.

It's really all about figuring out what are the most likely mechanisms that we're going to find natural language syntax in the brain, which parts of the brain are going to be sensitive to it and which really index that process on some level.

So, yeah, I think I guess I should probably leave it there and conclude it.

But, yeah, that's pretty much the whole model.

Thank you for the massively informative overview and the research agenda.

Do you want to stay?

Happy to stay for a bit longer for sure.

Yeah, yeah.

I'm happy to chat.

All right.

Well, there's so many

places to join.

For people who are watching live, maybe we'll have a few minutes to ask a question.

Wow.

Yeah.

Yeah.

If you have any questions or comments or thoughts, I mean, I can, I can, I can say here all day and just talk forever about this stuff, but I probably shouldn't.

All right.

I have a handful.

Let's do some short answers and just some general, because on the evidence and on the specific brain regions and specific timescales, you've made that abundantly clear.

So I'm going to ask some alternate questions.

So how do we study these densely woven cognitive phenomena without, how do we hold that tension without just resorting simply only always to holism or reductionism?

You kind of talked
about that with the
inside out and outside
in views, but how can
we approach a different
cognitive system?

Maybe not the human
brain and language,
language where even
the relationships
amongst the different
phenomena might be
themselves not known.

Okay.

I see what you mean.

So if there's some
kind of, you mean a
human specific cognitive
process that is not
language.

Yeah.

If you want to pull
out a different thing
or study a different
system, what kind of
account are we looking
for?

How are we going to
know we're on the
path that's not being
just massively misleading
and so on?

No, no, that's a good
question.

So I think, I think, so
the Rose architecture I
think is, is partially
applicable to lots of

different cognitive domains.

I think for attention and working memory, you know, face perception, olfaction, they all involve something at the R to O level, the coordination of spike timing and the, the realization of some kind of sensory motor transformations at the broadband gamma level.

The, the thing where we take over is really at the S level where you get the particular type of low frequency coordination that seems to be specific for language, but it's, you know, language is not the only thing that recruits, um, you know, low frequency, um, oscillatory phase codes to conduct the orchestra of, you know, representations that need to be externalized or protected.

Um, so I think the, it's, it's transferable in many ways.

The real, um, testable, uh, issue will be to do with whether or not these, um, you know, whether or not I'm correct that these different levels of linguistic structure are found only at

these levels, or maybe they're found across all of them.

Um, so for example, if effects of syntax specific responses can be found truly at the R, the O, the S and the E scales that I've mentioned, that muddies the waters because it means that, um, the, you know, the brain at every scale across all areas cares about the most niche specific issues in syntax.

We might have to turn ultimately to, um, state space architectures, um, which I suspect will be used to supplement rows, at least at the, at the, um, the, uh, the SME levels, I think in terms of you'll have like, um, different, um, abstract states being coordinated across network nodes, um, rather than the kind of, which is, you know, similar to what I mentioned to with, with, with traveling waves, just a more kind of global interactional process, but for a very different mathematical grounding, it'll have a very different mathematical grounding.

Um, so, but I think, um, I don't know, I don't, I'm not sure if I'm answering your question well, but I, I think,

I think in terms of transferring it to other domains that are, that could be explored, um, it's really about, um, you know, trying to find, trying to find a prediction of like, I will, I need to use a particular methodology.

I need to use a particular experimental paradigm to find effects that should only be found at this particular neural resolution.

And in this respect, it's no different from the rest of the cognitive sciences, because in working memory and attention and navigation, that's exactly what these guys have done for decades.

Uh, linguistics as usual is, you know, the last person to the party.

Um, we are usually, linguists are usually the last field in cognitive neuroscience to kind of, you know, pick up the developing tools in the, in the field and apply them.

Um, so, and that's fine.

You know, there's nothing wrong with that.

Uh, it turns, I think the reason is because human language is the most complex cognitive process that we have. It requires a lot of the weeds to be sorted out before we have to

figure out how working memory works, how attention works, how vision works before we can even worry about language.

Um, because all of them kind of presuppose and require that, right?

At some level.

So I think it's natural for language to be saved until last, right?

Save the best to last for sure.

No doubt.

Um, but I think it's also, it makes it, um, it's also the case that neuro linguists have not always taken the lessons that previous fields have learned as seriously.

And we often end up making the same mistakes.

So when we got scalp, scalp EEGs for the first time or, or fMRI, we made a lot of the same mistakes, um, that the, um, memory guys did or the navigation guys did.

Um, and I think my own fear, you know, in writing this paper is that the reason why I wrote this is mainly because, you know, I see it in the last few months, I see it in the last year or so, um, cognitive neuroscientists getting their hands on very advanced recording techniques, but not really knowing what to make of

it.

Again, this very aggressive, very, but also very contemporary and very progressive kind of so-called progressive viewpoint of, you know, data-driven science where you just kind of let the data speak for itself and, uh, allow the statistics of your, uh, you know, also do some kind of machine learning, um, on your, on your kind of, on your neural data and expect that to kind of feed into an explanation for language.

A lot of those techniques, um, also discourage this kind of theoretical reflection.

Um, and if you look at people like Carl Friston, you know, he spends most of his time, he's the world's most influential neuroscientist and he spends most of his time just, uh, doing theoretical neuroscience because it's, that's, that's really where we're at these days in a lot of these fields.

It's about consolidation and interpretation.

A lot of the, a lot of the, um, I mean, I was surprised when I, when I started writing this paper, I was surprised that a lot of the clues and answers to this puzzle of how syntax is inputting in the brain are kind of

already out there.

They just haven't been very, you know, uh, the dots haven't been connected.

the, uh, the principled relations and logical relations between different findings and different, um, you know, recording methodologies, um, and the kind of more principled relations between them haven't been established or haven't been figured out.

Um, and I think there's often a push and a desire to just, you know, do another experiment, uh, you know, get funding to do more and more experiments and collect more data and do more analysis.

That's just awfully essential.

I do that as my day, as my day job.

Um, well, outside of my day job, I think it's, it's also important to spend some time, you know, thinking about how to thinking more carefully about this so that you don't need to do any more experiments for a while.

Um, so you, or at least so you know that, you know, when you do your experiments, you know, the, the landscape better, um, and you don't end up making the same mistakes that other people do.

Um, but really it's, it's, it goes back to this issue of kind of cost, cost benefit.

Like, you know, how do you, how do you kind of weigh up the, um, the pros of collecting more data with the, um, you know, the, the, the foreseeable benefits book, the whole notion of foreseeable benefits, empirical foreseeable, foreseeable benefits really presupposes and needs this framework of what the benefits are likely to be.

If the benefits are just, I'm going to just let the data speak for itself and get a bunch of different, you know, machine learning, um, or LLM architectures to kind of figure out the story for me.

You're probably going to be disappointed.

Um, whereas if you take a step back and think about a more principled architectural approach, like think about the brain the way an engineer would, um, and figure out like which level is going to be best served for a different, a particular cognitive process.

That's, that's the basic mindset behind, behind this model.

It's about figuring out like what, you know, which scale of neural operation is going to be most helpful for you.

Great.

Well, on that kind of research program scale, the compass rose, it's a great model.

Um, I was curious about a few other settings for speech, like listening to a video or recording at a different speed than one X or watching subtitles or just the idea of reading and whether there was some kind of a, um, clean handoff between one sentence is heard and then one is read and then one is heard.

Is it beyond the sensory modality by what stage or what elements that are extra lexemic like prosody and timing and so on play into, for example, speech as opposed to writing where there's formatting and italics and so on.

Yeah, no, that's a good question.

So, um, there's a, there's an, there's an unanswered question at the moment, which is to do with the bottleneck of reading or speech process.

Well, speech processing, not too much, but definitely for reading.

Um, we don't really know that much about the bottleneck, like how fast you can read.

Um, so the Rose model talks about delta rhythms.

It talks about theta rhythms.

It's a very slow waves.

If you ever tried speed reading, it becomes very clear straight away that human beings can read very quickly.

Um, so if you, if you, you know, code in MATLAB or, or Python,

some kind of RSVP style, um, you know, uh, uh, a textbook where you can just like have one word flashing on the screen at a time and set it to different rates.

If you have, you know, 50, 500 milliseconds a word, no problem.

It turns out you can go way faster up to like, you know, 15, 20 words a second, and you can still get interpretability.

Um, you you'd be surprised if if you try it, you'll be very surprised by how fast you can actually read.

Um, so you, you might think that poses a problem for the Rose architecture.

I don't think it does.

I can get to that after I answer your question.

It doesn't pose a problem at all.

Um, but so that's, that's a good question.

So we know that there are certain parts of the brain that do, um, in the mental temporal cortex, um, which is the lower side of the other side of the temporal lobe.

They are responsible for, um, funneling, transforming visual information into orthographic information.

In fact, the work of Oscar Wall now in my lab, he's done the most impressive work on this by far.

Um, it's very exquisite, like intracranial recordings of how

human beings, um, in fact, his whole research is on reading. So you should probably definitely read his work.

It's basically about how the human brain takes, you know, visual information and then trans, uh, converts it into an orthographic space, a constrained orthographic space, a space in which, you know, uh, your, your mind, I guess, will be sensitive to particular statistical configurations of shapes, but you know, this one's a letter, this one's not a letter.

So that takes place in the same part of the ventral temporal cortex.

And then maybe somewhere around there that will, due to endogenous, um, you know, processing and limitations, probably not to do with vision, probably due to, you know, some kind of, uh, next, next stage, uh, process to do with interpretability.

Um, because as you know, we can read, we can see, uh, based on such the card rates and, and all the rest of the number of like objects and things we can see in a given second is, is, is, is very large indeed.

The, the human brain mostly filters out, you know, most of its sensory information every second, just to kind of focus on the stuff

that's key to its general model, the stuff that's key to kind of, um, you know, minimizing the free energy of its, of its general model and keeping it kind of sustained and, um, not changing the status quo, et cetera.

But on the reading level, uh, we know that, um, at that stage, that's when you get the sensory motor, what I call the sensory motor transformations in high gamma activity.

So that's where you get that high gamma activity being responsible and sensitive to the mapping from that visual information to the linguistic information, because obviously somehow you've got to, you see words on a screen, you see squiggles and lines, somehow you've got to convert that into a more abstract space.

So that all takes place in the ventral temporal cortex.

And then you get some interesting connectivity and top down information from being sent from frontal cortex back to ventral temporal to constrain and re and, and kind of coordinate the, that bottom there.

At least that's one of the main theories that that exists.

And I'm not an expert on reading per se.

I'm not a vision scientist, but, you know, as far as I understand, that's the kind of, um, the general

framework is that there is a part of the brain that's sensitive to the statistics, but also the symbology of orthographic information.

Um, um, and we know that it takes place in very short timescales.

Um, when it comes, you, you, you asked a very interesting question about the, the kind of, um, you know, co the kind of dual modality, like when you're watching a film, you're listening it and you're also reading subtitles at the same time.

Um, that's a very tricky question because, um, on the central motor side, we know that you'll get oratory and cortex involvement and orthographic kind of occipital temporal, um, involvement engagement at the same time, but somehow they both have to converge on a single interpretation, right?

Because we know that, um, you can't comprehend things in parallel.

You can't read two books at once.

Um, or at least most people can't.

So that means there's got to be a single kind of interpretation procedure and a single kind of semantic integration process that is being instructed in a form by different modalities.

But when you think about it, that might sound difficult, but it's actually no different from when I use hand gestures right now, right?

So when I use hand gestures, I'm providing some kind of extra modal

information to you.

Um, it's obviously easier because this is more hand gestures are usually kind of pragmatically informative rather than explicitly propositional, of course, unless you speak sign language.

Um, but with, it's, it's still just a case of like more different sensory motor transformations taking place. So that would still be at the art of the old level.

That would still be at the, um, the old level of art.

It's still high gamma transformations that are then fed into a more uniform, um, uh, discrete and abstract syntactic inference being made.

There's only one syntactic inference being made about the meaning of the sentence, but it's being informed by multiple sites.

Now you do raise an interesting question because that might, that might be the case.

It might be the case, therefore, that this low frequency delta complex therefore has to entrain and functionally connect with multiple cortical sites.

It might have to speak to auditory cortex, but also ventral temporal cortex.

If you're, if you're gleaning distinct, um, types of, um, if, if, if, if, if the lower level lexical representations you're getting are

being sourced from auditory, that orthographic.

So for example, if I give you a sentence and I chop and I get you to hear the first word and then read the second word and then hear, read, hear, read, hear, or listen, I should say, listen, read, listen, read.

If you go between listening and reading, but you're still integrating a single unification, a unified structure, we know you can do that. Human beings can do that.

So that tells you straight away that, you know, it can be done.

Um, but the question is how it's done.

Like I said, I suspect it's through this, this dual kind of trade-off between central motor, um, instructions being sent from auditory cortex to, to the same unified low frequency program.

And then the visual information too.

So that's, that's actually a kind of a nice way to separate out the distinction between, you know, central motor stuff from the more abstract unified kind of syntactic information that's being built.

Um, and in fact, you've just given me a good idea for an experiment.

So thank you.

Cool.

Yeah.

Because language is so linearized.

It's like being pulled through a

needle and then in the saccade so called speed reading, but it's not simply moving the linearization faster.

Speed reading can leverage formatting and all these other strategies, like almost thinking through other minds ways of reading and also having an off center effect in visual system.

And the way that that kind of convolves the orthography versus in listening, you basically always want to be listening more clearly because the moment is the only time you're going to hear it.

The active listening components, like you can't let it fly by, but reading you can, you can take secade strategies and cognitive strategies that, that you just can't take.

And that's kind of this like all at once versus one at a time element.

There's actually, there's a framework in, in psych linguistics called, um, uh, the now or never bottleneck.

So the now or never bottleneck refers to what you're saying, right?

In, in reading, there's no now or never.

You can just take your time with listening.

You're totally right.

It's, it's now or never you have to pay attention now and integrate or your working memory is going to dissipate or your attention is going to, or your phonological loop is going to decay.

And you're going to ask the person to

repeat themselves again and again.

Um, so it's a very, it's a very
different performance system.

And it's also very different from
braille.

Um, when, when you're touching, using
touch and you're reading braille, when
you're gesturing and using sign
language, you know, language, especially
syntax truly is a mobile.

It's, it's completely independent of any
modality.

Um, you know, the neuroscience of speech
is very different from the neuroscience of
language.

The neuroscience of, of gesture is very
different from neuroscience.

Your language is a more kind of a
modal abstract system that calls upon
these very various sensory motor kind of
processes.

Um, which is what I'm, you know, most
people intuitively trying ground language
in a sensory motor kind of context, because
that's natural, of course, you know, that
that's how we access it.

That's how we trigger it.

Well, obviously, you know, most of, most
of our daily language use is probably just
thinking to ourselves, you know, um, using
language, maybe subconsciously using the,
using the computational capacity that language
affords us in non-linguistic ways, meaning
to, to construct hierarchical structures of
thought and planning and, uh, you know, uh, and
all the rest of it.

Um, in, in some kind of subconscious way, uh,
rather than explicitly saying, I am saying a

sentence or even I am thinking a sentence in my head, because even when we think a sentence in our head, that's still just an internalized form of externalization, right?

So when you think about in your head, that's still, that's still externalization, which is a different type of externalization.

You're not externalizing it with your mouth, but you're still putting it out into your own phonological loop, but that's not true.

That's still externalization.

Internalization is truly the subconscious process of generating the syntactic inference, which is done rapidly, reflexively, uh, you know, intuitively, often without much thought, often without meaning attention, often without much attention being directed to it and any kind of conceptual implications being made.

Um, so most of language use is outside of the sensory motor space.

I would argue, maybe I'm wrong about that, but, um, this is a point that Noam Chomsky has made a lot, and I think he's right about this.

Um, you know, a lot of the use of like the daily use of the language infrastructure, um, is, like I said, I mentioned the universal currency metaphor.

It's like, it's across these different conceptual spaces.

Um, and it's, it's, it's, it's potentially represented very redundantly across sparse neural codes, as I mentioned in, in the Rose architecture, such that it survives brain damage.

It survives all sorts of different things.

Um, but, um, but it's still, it's still crucial.

And you get a lot of people like Rosemary Barley and Federico, you know, going and showing work that, um, you know, damage to,

well, you know, using imaging research, showing that, um, the language network is not activated during XYZ non-linguistic cognitive tasks or damage to the language network does not damage, you know, XYZ non-linguistic tasks.

And that's all fine.

Um, but it's really, that's kind of a separate issue from whether or not the more abstract syntactic neural code that I've presented in the paper is recruited on some level by non-linguistic, uh, resources, which I suspect it is.

Um, again, another empirical question for you.

Right.

Um, but I think that's a reasonable framework to kind of assume at least.

Uh, yeah.

Cool.

One, one, I guess, reflection on that.

And then one last question.

Um, you described language, human language as being a kind of psycho technology with supporting architecture and enculturation, all these different features that are, are totally relevant, influence it, that allows us to articulate like the sense making in action.

Like when we look at figure 4.3 in the active inference textbook, and we have a Bayesian graph that factorizes out through the sparsity of just how things are related, like how the measurement of the thermometer is related to the temperature in the room, how the temperature in the room changes, fundamental concepts of place and time, where preferences fit into that, what this is about, the semantics of this variable, whether it's kind of like a central tendency or whether it's a variance estimator like confidence, which is something that you brought up as like a strong linguistic, um, utility.

So it's kind of like it has an element of just being the

geometry and topology of thought and thinking and generative modeling that has ultra mundane everyday uses because this generalized apparatus really does have utility, pragmatic value in communication.

So it can be used for the most referential, um, just one bump above pointing at something and pointing at something and grunting and pointing at something and say, look, or look, it's a, like you can keep going.

And, but often what's only needed is the first little bit, especially given a context.

And yet there's this open-ended arbitrary state space navigation elements as well.

Yet it's only, I just, I think it's incredible how it's laid out and how what you described with a heavy emphasis on the neurophysiological and the computational comes into play with active inference and the generalized state spaces and all of these concepts.

So that's just super exciting.

I guess my sort of closing or, or we can keep talking or whatever, but how do you see this relating to language learning or first language and for not first language learning?

Yeah.

I, I, I, I totally agree with what you said about the active inference framework, but I think, I think, um, you know, linguistic semantics and well, you know, syntactic structures are a unique kind of inference that the human brain has to make.

Um, the, um, I, I, I see a lot of the, the properties of lexical semantics in human language as being a unique type, a unique contributor to the active inference framework.

Um, and I think it's, it's a surprise to me that more linguists haven't turned to active inference to think about certain properties of some, cause it seems obvious. It seems so compatible.

Uh, I've tried to do it in some work.

Um, and I think maybe some other people have to potentially, but it's, um, it's, it's an obvious kind of, well, I mean, I've cited the Peter Hago paper.

He cites that way too, a little bit.

Um, but no, it's, it's a very promising way to, to pursue it.

Um, in terms of, um, language learning and acquisition, that's a very, very good question.

So in the, I mentioned that at the beginning, some of the low frequency, um, Delta responses

to syntax, all of that research is in adults.

Well, all of it was until, um, some researchers at Cambridge, UK, a few years ago started to think, okay, well maybe is this how children process language too?

Um, well maybe is this how, is this where they process language in the same regions?

It turns out if you look at, um, teenagers and children and you do the same kind of loaf, um, you know, scalp BG or maybe even Meg research.

I'm not sure.

I can't remember, I cited it in my book.

Um, the, uh, low frequency responses change over development in terms of which, um, ranges are sensitive to the periods of syntax.

So it's not necessarily, you know, 1.3 Hertz precisely that entrains the syntax.

Of course not.

This gets back to the point I made earlier about speed reading, right?

It's all about the flexible coordination and packaging of representations, um, into a given complex.

Um, again, there's, I mentioned the issue of trade-off between, you know, number of representations you can chunk and the, and the fidelity of the representations.

You can have a few and what represented in Ghana or a large number and only a small number of spikes, uh, per representation.

That's just kind of, you know, that's a very general framework that's been, um, supported in the attention literature too.

It's not, it's not just, um, my, my proposal.

So, so that, that, that poses a question.

If children and teenagers show a different kind of low frequency phrase response, how do we deal with that?

Um, I don't have a clear answer to that question.

I think the Rose architecture can very readily be modified to that because it's not as if, you know, for children, they'll still have the same architecture.

They'll still have the same, you know, R to O to S to E. The particular frequency band of interest might shift over development.

And in fact, we know from working memory and attention that theta dynamics and alpha dynamics definitely, and beta dynamics too, do change cross cortically over, over development.

Um, and these are behaviorally relevant to things like, um, development of working memory and attentional resources.

That's, that's been very clearly studied and very well studied too.

Um, so it's more like the, there's a more abstract code that simply shifts its spatial temporal profile as the brain gets older and then changes more when we get even older.

So I have a paper that's going to, um, uh, come out fairly soon, I hope to do with, um, aging.

Um, what happens when we, when our brains decay and we get old, um, there's, there's some nascent kind of research looking at the, uh, language processing, not at the behavior level, but in terms of the neural level, the neurobiology of, um, language processing in older people, um, and how it might also relate to, you know, breakdown of semantic information access and things like that.

And what are the potential correlates of that?

So you can imagine very clearly if you take the Rose architecture, which components will be disrupted or modified based on if it's a syntactic or a conceptual or a lexic critical kind of deficit, but you know, the, the, the, the isomorphism is, as I said, very crude. So you can, you can predict that very clearly.

And in fact, there's some research in that space that I think is concordant with this, which is what I discussed in this new paper.

Um, so, um, it's, it's, you know, it, it, like I said, it's a relevant architecture.

It's a very difficult question about how the whole brain, um, matures over time because the, you know, the certain aspects of knowledge that may not be represented, um, in ways that we currently know how to, um, you know, uh, detect or record.

Um, we have the, the, the, the four or five measures, main measures that I mentioned in the paper, they're, they're what we have right now.

Maybe there's a particular form of neural encoding that it just goes way beyond what we can physically provide in the year 2023.

Um, I don't think there's any reason to assume that we've reached the limit of like decodability of neural signals and their complexity, absolutely not.

Um, so the learning process is, is a very mysterious one.

Um, so yeah, I think, I think it's, it's also, it's also a really good way to falsify a lot of this stuff, by the way, it's a really good way just to completely falsify and test a lot of these proposals to do with the architecture itself.

Um, and then the developing code over time as well.

So yeah, it's a very relevant concern.

The, the, the best way to do it is, um, to get inter-trainer recordings and children.

Uh, that's very difficult because, so there's a, uh, in Austin, Texas, there's a lab there that has, that has, that has these recordings and, and uses, um, access to, uh, uh, children with epilepsy.

And they get these children to, you know, watch trailers of films and listen to stories and things like that.

And they do some interesting language research, uh, in these kids.

That's exceptionally rare data and very, very, very useful data for answering these questions.

I don't have access to that data.

Um, and most of that research has not been forthcoming yet.

There's, I, there's been a couple of intracranial, um, papers in, in, in children and language, but it's, it's fairly sparse, fairly low patient numbers.

And the types of analyses have been too kind of remote and departed from this framework to really test it.

But that would be the way to test it.

Yeah.

You know, um, electrodes directly inside the brains of, of, of children and potentially even younger.

I think the youngest I've seen is about four.

Um, but you know, even by four years old, it's maybe too late to, to chart the actual full development of acquisition of language.

Cause by that stage, they've already acquired syntax, right?

If you could get even, I don't want to sound like a mad scientist, but like if you, if it's possible ethical to, to one day get access to intracranial recordings and very young children, then it would be possible to chart this developmental process, especially if the children are in the hospital for, uh, protracted periods.

And if I get some kind of, I mentioned BCI implants, if they have some kind of, uh, implanted device over time, that would be the most direct way to test this stuff.

And well, not just test my theory, but just, you know, just, just completely chart out the full developmental profile and the evolution and the development of the neurophysiological basis of speech processing and language processing.

Um, but right now that's kind of, as far as I understand, that's kind of like the current landscape of, of, of, of where the field is right now.

Wow.

Well, um, you can have the last word with any last thoughts, but thank you for this amazing presentation.

And I hope that if people are interested that they can of course, read the paper and learn and reach out and maybe we can have 64.2 when the time is right, but anything you'd like to kind of leave it with.

Yeah, no, thanks so much for having me on that.

Um, it's been very, it's been great.

You've been great.

I've, I've been enjoyed.

It's been enjoyable just kind of outlining my own theory.

You know, you read your own paper over and over again and you realize all the things that you kind of misinterpreted or misread on your face, read through, um, in terms of like phrasing and all the rest of it.

So it's always good to, to refresh these things, but no, absolutely.

If people are interested in, in this stuff, I'm always open.

And, you know, you can contact me every email, uh, on, on Twitter.

Um, very happy to talk more.

Absolutely.

So yeah, thank you for listening.

Thank you for having me.

Thank you.

Peace.

Peace.

Peace.

Thank you.